E17e - Fourier Analysis of Coupled Electric Oscillations

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1 Introduction

Coupled oscillating systems occur in many areas of physics, from mechanical pendula to electromagnetic resonators. In this experiment, coupled electrical oscillations are investigated using two resonant LC circuits. A single LC circuit oscillates at its natural frequency, which is determined by its inductance and capacitance. When two such circuits are coupled, energy can be transferred periodically between them. As a result, the original resonance splits into two normal modes: an in-phase mode, in which both circuits oscillate together, and an out-of-phase mode, in which the two circuits oscillate with a phase difference of 180°. If both normal modes are excited simultaneously, the superposition produces a beat oscillation in the time domain.

The aim of this experiment is to study these coupled oscillations using both time traces and frequency spectra. The time-dependent voltage signals are recorded with a USB oscilloscope, and their frequency components are obtained by Fourier transformation. The peaks in the frequency spectra allow the normal-mode frequencies to be determined. From these frequencies, the coupling factor of the system can be calculated and compared with theoretical predictions.

The experiment consists of four main parts. First, the low-point circuit of two capacitively coupled LC circuits is investigated for different values of the coupling capacitance C_K. For each coupling capacitance, the free oscillation and its frequency spectrum are measured, and the frequencies of the in-phase and out-of-phase modes are determined. The corresponding coupling factors are then calculated and fitted with the theoretical expression in order to determine the capacitance C of the resonant circuits.

Second, the high-point capacitively coupled circuit is studied. In this configuration, the initial charging conditions allow the in-phase and out-of-phase modes to be excited more directly. As in the low-point circuit, the measured mode frequencies are used to calculate the coupling factors for several values of C_K, and the theoretical dependence is fitted to obtain another estimate of the capacitance C.

Third, a beat oscillation is measured for one selected coupling capacitance in the high-point circuit. The beat period is determined directly from the time trace and compared with the value calculated from the difference between the two normal-mode frequencies. This provides a direct connection between the time-domain behaviour and the frequency-domain spectrum.

Finally, inductively coupled resonant circuits are investigated using two flat coils placed at different distances from each other. The frequency splitting is measured as a function of coil separation, and the corresponding coupling factor is calculated. From the uncoupled resonance frequency, the self-inductance of the coil can be determined, allowing the mutual inductance to be obtained. The distance dependence of the inductive coupling is then analyzed and compared with the expected decrease of the magnetic field along the coil axis.

Throughout the report, measurement uncertainties are included in the analysis. The main sources of uncertainty are the finite frequency resolution of the Fourier spectra, the identification of spectral peak positions, component tolerances, damping of the oscillations, and geometrical uncertainties in the coil-distance measurement. These uncertainties are propagated to the calculated coupling factors, fitted capacitances, beat periods, and mutual inductances, so that the agreement between measurement and theory can be assessed quantitatively.

2 Theoretical Analysis

2.1 Overview

The experiment investigates the normal modes of two coupled electrical resonant circuits. An individual resonant circuit consists of an inductance L and a capacitance C. If two identical resonant circuits are coupled, the original resonance frequency splits into two normal-mode frequencies. These two modes are the in-phase mode and the out-of-phase mode. In the in-phase mode, the currents or voltages in the two circuits oscillate with the same phase. In the out-of-phase mode, they oscillate with a phase difference of 180°.

The measured voltage signal $U(t)$ is recorded in the time domain using an oscilloscope. The corresponding frequency spectrum is obtained by Fourier transformation. The normal-mode frequencies are identified from the peak positions in the frequency spectrum. From these frequencies, the coupling factor can be calculated. For capacitive coupling, the coupling factor is then fitted as a function of the coupling capacitance C_K in order to determine the capacitance C of the resonant circuits. For inductive coupling, the frequency splitting is used to determine the coupling factor k_L and the mutual inductance M as a function of the coil distance.

Throughout the following derivations, the two resonant circuits are assumed to be identical,

$$C_A = C_B = C, \qquad L_A = L_B = L, \tag{1}$$

and damping is neglected when deriving the eigenfrequencies. In the real experiment, damping is present because of ohmic losses in the coils, wires, and other circuit components. This damping mainly causes the measured time traces to decay and the spectral peaks to broaden, but for weak damping it does not significantly change the resonance frequencies.

2.2 Single damped LC resonant circuit

Before considering coupled circuits, the resonance frequency of a single LC circuit is required. For a series RLC circuit, Kirchhoff's voltage law gives

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = 0, \tag{2}$$

where $q(t)$ is the charge on the capacitor and R represents the total resistance of the circuit. Dividing by L gives

$$\frac{d^2q}{dt^2} + \frac{R}{L}\frac{dq}{dt} + \frac{1}{LC}q = 0. \tag{3}$$

Introducing

$$\beta = \frac{R}{2L}, \qquad \omega_0 = \frac{1}{\sqrt{LC}}, \tag{4}$$

the differential equation becomes

$$\frac{d^2q}{dt^2} + 2\beta\frac{dq}{dt} + \omega_0^2 q = 0. \tag{5}$$

For weak damping, $\beta < \omega_0$, the solution is

$$q(t) = q_0 e^{-\beta t}\cos(\omega_d t + \phi), \tag{6}$$

with the damped angular frequency

$$\omega_d = \sqrt{\omega_0^2 - \beta^2}. \tag{7}$$

If the damping is small, $\beta \ll \omega_0$, then

$$\omega_d \approx \omega_0 = \frac{1}{\sqrt{LC}}. \tag{8}$$

Therefore, the natural frequency of an ideal LC circuit is

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi\sqrt{LC}}. \tag{9}$$

2.3 Fourier analysis and frequency spectra

The oscilloscope records the voltage as a function of time, $U(t)$. The frequency spectrum is obtained by Fourier transformation. The continuous Fourier transform may be written as

$$\tilde{U}(f) = \int_{-\infty}^{\infty} U(t)e^{-i2\pi ft}\,dt. \tag{10}$$

In the experiment, the measured signal is finite and discrete, so the oscilloscope uses a fast Fourier transform (FFT). If the total recorded time interval is T_{rec}, the frequency spacing of the FFT spectrum is approximately

$$\Delta f = \frac{1}{T_{\mathrm{rec}}}. \tag{11}$$

This frequency spacing limits how precisely the peak positions can be determined.

If the sampling frequency is f_s, the largest frequency that can be represented without aliasing is the Nyquist frequency,

$$f_{\mathrm{Nyquist}} = \frac{f_s}{2}. \tag{12}$$

Frequencies above f_{Nyquist} would be incorrectly represented as lower frequencies in the measured spectrum. Therefore, the sampling frequency must be chosen high enough that the relevant resonance frequencies lie well below the Nyquist frequency.

2.4 General coupling factor from frequency splitting

For two coupled resonant circuits, the coupling causes a splitting of the resonance into two angular frequencies, denoted here by ω_1 and ω_2. The coupling factor is defined by

$$k = \left| \frac{\omega_1^2 - \omega_2^2}{\omega_1^2 + \omega_2^2} \right|. \tag{13}$$

Since

$$\omega = 2\pi f, \tag{14}$$

the factor $(2\pi)^2$ cancels in the ratio. Thus, the experimentally used expression is

$$k = \left| \frac{f_1^2 - f_2^2}{f_1^2 + f_2^2} \right|. \tag{15}$$

This equation is used to calculate the coupling factor directly from the measured peak frequencies in the FFT spectra.

2.5 Capacitive coupling: low-point circuit

In the low-point circuit, the two LC resonators are coupled through the coupling capacitor C_K. Let I_A and I_B be the currents in circuits A and B. For identical circuits and negligible damping, Kirchhoff's laws lead to the coupled differential equations

$$L\frac{d^2 I_A}{dt^2} + \frac{1}{C} I_A + \frac{1}{C_K}(I_A - I_B) = 0, \tag{16}$$

$$L\frac{d^2 I_B}{dt^2} + \frac{1}{C} I_B - \frac{1}{C_K}(I_A - I_B) = 0. \tag{17}$$

To find the normal modes, the equations are added and subtracted. Adding Eqs. (16) and (17) gives

$$L\frac{d^2}{dt^2}(I_A + I_B) + \frac{1}{C}(I_A + I_B) = 0. \tag{18}$$

This is the equation of a simple harmonic oscillator with angular frequency

$$\omega_1 = \omega_0 = \frac{1}{\sqrt{LC}}. \tag{19}$$

This mode corresponds to the in-phase oscillation, because the sum $I_A + I_B$ is non-zero while the difference is not dominant.

Subtracting Eq. (17) from Eq. (16) gives

$$L\frac{d^2}{dt^2}(I_A - I_B) + \left(\frac{1}{C} + \frac{2}{C_K}\right)(I_A - I_B) = 0. \tag{20}$$

Therefore, the second normal-mode angular frequency is

$$\omega_2 = \sqrt{\frac{1}{L}\left(\frac{1}{C} + \frac{2}{C_K}\right)}. \tag{21}$$

This mode corresponds to the out-of-phase oscillation.

The total current in either circuit can be written as a superposition of the two normal modes:

$$I_A(t) = \frac{1}{2}(I_{A,0} + I_{B,0})\cos(\omega_1 t) + \frac{1}{2}(I_{A,0} - I_{B,0})\cos(\omega_2 t), \tag{22}$$

$$I_B(t) = \frac{1}{2}(I_{A,0} + I_{B,0})\cos(\omega_1 t) - \frac{1}{2}(I_{A,0} - I_{B,0})\cos(\omega_2 t). \tag{23}$$

Thus, if only one circuit is initially excited, both normal frequencies appear in the measured signal, producing a beat-like free oscillation.

Using the general definition of the coupling factor,

$$k_{C,T} = \left| \frac{\omega_1^2 - \omega_2^2}{\omega_1^2 + \omega_2^2} \right|, \tag{24}$$

and substituting Eqs. (19) and (21), we obtain

$$\omega_1^2 = \frac{1}{LC}, \tag{25}$$

$$\omega_2^2 = \frac{1}{L}\left(\frac{1}{C} + \frac{2}{C_K}\right). \tag{26}$$

Therefore,

$$k_{C,T} = \frac{\frac{2}{LC_K}}{\frac{2}{LC} + \frac{2}{LC_K}} \tag{27}$$

$$= \frac{\frac{1}{C_K}}{\frac{1}{C} + \frac{1}{C_K}} \tag{28}$$

$$= \frac{C}{C + C_K}. \tag{29}$$

Equation (29) is the theoretical fit function for Task 1.

2.6 Capacitive coupling: high-point circuit

In the high-point circuit, the coupling capacitor C_K connects the upper points of the two resonant circuits. This arrangement makes it possible to define the initial conditions more clearly. The in-phase mode is excited by charging both circuits with the same voltage,

$$U_a = U_b, \tag{30}$$

whereas the out-of-phase mode is excited by charging the two circuits with opposite voltages,

$$U_a = -U_b. \tag{31}$$

Let $I_{C,A}$ and $I_{C,B}$ be the capacitor currents in the two circuits. The normal coordinates are defined as

$$I_+ = I_{C,A} + I_{C,B}, \qquad I_- = I_{C,A} - I_{C,B}. \tag{32}$$

After applying Kirchhoff's laws and using these normal coordinates, the coupled equations decouple into

$$\frac{I_+}{C} + L\left(1 + \frac{2C_K}{C}\right)\frac{d^2 I_+}{dt^2} = 0, \tag{33}$$

and

$$\frac{I_-}{C} + L\frac{d^2 I_-}{dt^2} = 0. \tag{34}$$

Equation (33) gives

$$\omega_+ = \frac{1}{\sqrt{L(C + 2C_K)}}. \tag{35}$$

Equation (34) gives

$$\omega_- = \frac{1}{\sqrt{LC}}. \tag{36}$$

Since $C_K > 0$, it follows that

$$\omega_- > \omega_+. \tag{37}$$

The current can be written as a linear combination of the two modes:

$$I(t) = A_+ \cos(\omega_+ t) + A_- \cos(\omega_- t), \tag{38}$$

where the amplitudes A_+ and A_- depend on the initial conditions.

The high-point capacitive coupling factor is

$$k_{C,H} = \frac{\omega_-^2 - \omega_+^2}{\omega_-^2 + \omega_+^2}. \tag{39}$$

Using

$$\omega_-^2 = \frac{1}{LC}, \qquad \omega_+^2 = \frac{1}{L(C + 2C_K)}, \tag{40}$$

we obtain

$$k_{C,H} = \frac{\frac{1}{LC} - \frac{1}{L(C+2C_K)}}{\frac{1}{LC} + \frac{1}{L(C+2C_K)}} \tag{41}$$

$$= \frac{\frac{2C_K}{LC(C+2C_K)}}{\frac{2(C+C_K)}{LC(C+2C_K)}} \tag{42}$$

$$= \frac{C_K}{C+C_K}. \tag{43}$$

Equation (43) is the theoretical fit function for Task 2.

2.7 Beat oscillations

A beat oscillation occurs when both normal modes are excited simultaneously. For simplicity, consider two oscillations with equal amplitudes:

$$U(t) = U_0 \cos(\omega_1 t) + U_0 \cos(\omega_2 t). \tag{44}$$

Using the trigonometric identity

$$\cos a + \cos b = 2\cos\left(\frac{a-b}{2}\right)\cos\left(\frac{a+b}{2}\right), \tag{45}$$

the signal becomes

$$U(t) = 2U_0 \cos\left(\frac{\omega_2-\omega_1}{2}t\right)\cos\left(\frac{\omega_1+\omega_2}{2}t\right). \tag{46}$$

The second cosine describes the fast oscillation with average angular frequency

$$\bar{\omega} = \frac{\omega_1+\omega_2}{2}. \tag{47}$$

The corresponding oscillation period is

$$T = \frac{2\pi}{\bar{\omega}} = \frac{4\pi}{\omega_1+\omega_2}. \tag{48}$$

Using $\omega = 2\pi f$, this becomes

$$T = \frac{2}{f_1+f_2}. \tag{49}$$

The slowly varying envelope is determined by the frequency difference. The beat angular frequency is

$$\omega_S = |\omega_2 - \omega_1|. \tag{50}$$

Therefore, the beat period is

$$T_S = \frac{2\pi}{|\omega_2-\omega_1|} = \frac{1}{|f_2-f_1|}. \tag{51}$$

In Task 3, Eq. (51) is used to calculate the beat period from the two spectral peak frequencies and compare it with the beat period measured directly from the time trace.

2.8 Inductive coupling

In Task 4, the two resonant circuits are coupled inductively. The magnetic field produced by one coil induces a voltage in the other coil. This effect is described by the mutual inductance M. For two identical resonant circuits, Kirchhoff's laws give

$$\frac{I_A}{C} + L\frac{d^2I_A}{dt^2} + M\frac{d^2I_B}{dt^2} = 0, \tag{52}$$

$$\frac{I_B}{C} + L\frac{d^2I_B}{dt^2} + M\frac{d^2I_A}{dt^2} = 0. \tag{53}$$

As before, the normal coordinates are

$$I_+ = I_A + I_B, \qquad I_- = I_A - I_B. \tag{54}$$

Adding Eqs. (52) and (53) gives

$$\frac{I_+}{C} + (L+M)\frac{d^2I_+}{dt^2} = 0. \tag{55}$$

Thus,

$$\omega_+ = \frac{1}{\sqrt{C(L+M)}}. \tag{56}$$

Subtracting the two equations gives

$$\frac{I_-}{C} + (L - M)\frac{d^2 I_-}{dt^2} = 0. \tag{57}$$

Thus,

$$\omega_- = \frac{1}{\sqrt{C(L-M)}}. \tag{58}$$

The inductive coupling factor is defined as

$$k_L = \frac{M}{L}. \tag{59}$$

Using the uncoupled angular frequency

$$\omega_0 = \frac{1}{\sqrt{LC}}, \tag{60}$$

Eqs. (56) and (58) can be written as

$$\omega_\pm = \frac{\omega_0}{\sqrt{1 \pm k_L}}. \tag{61}$$

For weak coupling, $k_L \ll 1$, the frequency splitting is approximately

$$\Delta\omega = \omega_- - \omega_+ \approx k_L \omega_0. \tag{62}$$

The exact coupling factor can be obtained from the measured split frequencies:

$$k_L = \frac{\omega_-^2 - \omega_+^2}{\omega_-^2 + \omega_+^2}. \tag{63}$$

Since $\omega = 2\pi f$, this becomes

$$k_L = \frac{f_-^2 - f_+^2}{f_-^2 + f_+^2}. \tag{64}$$

The self-inductance L of one coil can be determined from the resonance frequency f_0 of the uncoupled LC circuit. From Eq. (9),

$$f_0 = \frac{1}{2\pi\sqrt{LC}}, \tag{65}$$

so that

$$L = \frac{1}{(2\pi f_0)^2 C}. \tag{66}$$

Once L and k_L are known, the mutual inductance is

$$M = k_L L. \tag{67}$$

2.9 Distance dependence of the mutual inductance

In the inductive-coupling part of the experiment, the coil distance d is varied. As the distance increases, the magnetic flux from one coil through the other decreases, so the mutual inductance M and the coupling factor k_L are expected to decrease.

The measured distance dependence is described by a power law,

$$M(d) = A d^{-n}, \tag{68}$$

where A is a proportionality constant and n is the power-law exponent. Taking the logarithm of both sides gives

$$\ln M = \ln A - n \ln d. \tag{69}$$

Thus, a plot of $\ln M$ against $\ln d$ should be approximately linear if the power law is valid. The slope of the line is $-n$. For a circular coil, the magnetic field on the symmetry axis decreases with distance according to

$$B(d) = \frac{\mu_0 N I R^2}{2(R^2 + d^2)^{3/2}}, \tag{70}$$

where R is the coil radius and N is the number of turns. For distances much larger than the coil radius, $d \gg R$, this becomes approximately

$$B(d) \propto d^{-3}. \tag{71}$$

Since the mutual inductance is proportional to the magnetic flux through the second coil per current in the first coil, one expects approximately

$$M(d) \propto d^{-3} \tag{72}$$

in the far-field limit. However, for distances comparable to the coil radius, deviations from a pure d^{-3} dependence are expected. Therefore, the fitted exponent n is compared with the value $n = 3$ only as an approximate physical reference.

2.10 Summary of theoretical expressions used for the analysis

The main equations used in the data analysis are summarized below.

For the uncoupled LC circuit,

$$f_0 = \frac{1}{2\pi\sqrt{LC}}. \tag{73}$$

For the coupling factor calculated from two measured frequencies,

$$k = \left| \frac{f_1^2 - f_2^2}{f_1^2 + f_2^2} \right|. \tag{74}$$

For the low-point capacitive circuit,

$$k_{C,T} = \frac{C}{C + C_K}. \tag{75}$$

For the high-point capacitive circuit,

$$k_{C,H} = \frac{C_K}{C + C_K}. \tag{76}$$

For the beat oscillation,

$$T_S = \frac{1}{|f_2 - f_1|}, \qquad T = \frac{2}{f_1 + f_2}. \tag{77}$$

For the inductively coupled circuit,

$$k_L = \frac{f_-^2 - f_+^2}{f_-^2 + f_+^2}, \qquad L = \frac{1}{(2\pi f_0)^2 C}, \qquad M = k_L L. \tag{78}$$

For the distance dependence of mutual inductance,

$$M(d) = A d^{-n}. \tag{79}$$

3 Experimental Analysis

3.1 Experimental Setup

The experiment was performed using two coupled electrical resonant circuits. Each resonant circuit consisted of a capacitor C and an inductor L. Depending on the task, the two circuits were coupled either capacitively by a variable coupling capacitor C_K or inductively by the magnetic field of two facing coils.

The voltage signals were measured with a USB oscilloscope connected to a computer. The oscilloscope was operated in time-domain and spectrum modes, so that the voltage signal $U(t)$ and the corresponding frequency spectrum could be recorded simultaneously. The spectrum was obtained by the built-in fast Fourier transform (FFT) function of the oscilloscope. The oscilloscope input was connected across the capacitor of the measured resonant circuit. The ground connection of the oscilloscope was connected to the zero-potential socket of the switchbox.

The switchbox was used to charge and discharge the capacitors before starting the free oscillation. It provided the voltages $+10\,\mathrm{V}$, $-10\,\mathrm{V}$ and zero potential. The capacitors were first charged using the switchbox voltage, and the oscillation was then started by switching the circuit so that the stored electrical energy could oscillate freely between the capacitors and inductors. For the triggered measurements, the oscilloscope was set to single-shot mode and triggered by a descending voltage flank through $0\,\mathrm{V}$.

For the capacitively coupled circuits, the coupling capacitor C_K was provided by a capacitor decade. This allowed different values of C_K to be selected without changing the rest of the circuit. The same two LC circuits were used for the low-point and high-point coupling measurements. Before the coupled measurements were performed, both uncoupled resonant circuits were measured separately in order to compare their natural frequencies.

In the low-point circuit, the two LC series circuits were connected in parallel with the coupling capacitor C_K. The initial charging conditions in this configuration excite a mixture of the two normal modes, so that both resonance frequencies appear in the same spectrum.

In the high-point circuit, the coupling capacitor connected the upper points of the two resonant circuits. This setup allowed the initial charging states of the two capacitors to be chosen independently. The in-phase mode was excited by applying equal charging voltages, $U_a = U_b$, whereas the out-of-phase mode was excited by applying opposite charging voltages, $U_a = -U_b$.

For the beat measurement, the high-point circuit was used again. One of the circuits was shorted by connecting it to zero potential instead of the negative voltage supply. Only the other circuit was initially charged. After the switch was closed, energy was exchanged between the two circuits, producing a beat oscillation in the measured voltage signal.

For the inductive-coupling measurement, two flat circular coils were mounted facing each other on an optical bench. The distance d between the coils was varied systematically. The capacitors used in this part were the same as those used in the capacitively coupled circuits, so their capacitance values were known from the previous analysis. The coupling was caused by the mutual inductance M between the coils.

3.2 Data Acquisition and Measurement Procedure

The measured data consisted of voltage time traces and corresponding FFT spectra. For each measurement, the oscilloscope recorded the voltage across one capacitor as a function of time. At the same time, the FFT spectrum was displayed and saved. The positions of the spectral peaks were later used to determine the normal-mode frequencies of the coupled system.

Before the main measurements, the two uncoupled LC circuits were measured separately. For each circuit, the capacitor was charged with the dc voltage from the switchbox. After closing the switch, the capacitor discharged through the inductor and an exponentially damped free oscillation was observed. The resonance frequency of each circuit was determined from the peak in the FFT spectrum. This pre-measurement was used to check whether the two resonant circuits had approximately equal natural frequencies.

For Task 1, the low-point capacitively coupled circuit was assembled. The coupling capacitor C_K was set to ten different values using the capacitor decade. For each value of C_K, the capacitors were charged and the free oscillation was recorded after closing the switch. The time trace and frequency spectrum were saved for every measurement. Since the low-point circuit excites both normal modes simultaneously, two resonance peaks were expected in each FFT spectrum. The lower and higher peak frequencies were identified as the in-phase and out-of-phase mode frequencies, respectively. The measured frequencies were then used to calculate the coupling factor $k_{C,T}$ for each value of C_K.

The measurement procedure for Task 1 was therefore:

1. Assemble the low-point capacitively coupled circuit.
2. Select a value of the coupling capacitance C_K.
3. Charge the capacitors using the switchbox.
4. Start the free oscillation by closing the switch.
5. Record the voltage time trace and the FFT spectrum.
6. Determine the two peak frequencies f_1 and f_2 from the FFT spectrum.
7. Repeat the measurement for ten different values of C_K.

For Task 2, the high-point capacitively coupled circuit was assembled. Again, ten different values of C_K were used. For each value, the in-phase mode was excited by charging both circuits with the same voltage, $U_a = U_b$. The out-of-phase mode was then excited by charging the circuits with opposite voltages, $U_a = -U_b$. The corresponding FFT spectra were recorded, and the peak frequencies of the two normal modes were determined. These frequencies were used to calculate the coupling factor $k_{C,H}$.

The measurement procedure for Task 2 was:

1. Assemble the high-point capacitively coupled circuit.
2. Select a value of the coupling capacitance C_K.
3. Excite the in-phase and out-of-phase mode and record the spectrum.
4. Determine the corresponding peak frequencies f_1 and f_2.
5. Repeat this procedure for ten different values of C_K.

For Task 3, a beat oscillation was measured using the high-point circuit. One value of the coupling capacitance C_K was selected from the Task 2 measurements. One resonant circuit was initially charged, while the other circuit was shorted to zero potential. After the switch was closed, the voltage signal showed a fast oscillation with a slowly varying amplitude envelope. The oscillation period T and beat period T_S were determined directly from the time trace. The FFT spectrum of the same signal was also recorded, and the two peak frequencies were used to calculate the expected beat period for comparison with the directly measured value.

The measurement procedure for Task 3 was:

1. Use the high-point circuit with one selected value of C_K.
2. Short one of the two circuits to zero potential.
3. Charge the other circuit using the switchbox.
4. Start the free oscillation by closing the switch.
5. Record the beat signal in the time domain.
6. Record the corresponding FFT spectrum.
7. Determine T and T_S from the time trace.
8. Determine f_1 and f_2 from the FFT spectrum.

For Task 4, the capacitively coupled circuits were replaced by two inductively coupled resonant circuits. Two flat coils were positioned facing each other on an optical bench. The distance d between the coils was varied, and for each distance the free oscillation and FFT spectrum were recorded. The two split resonance frequencies f_+ and f_- were determined from the FFT spectrum. In addition, the uncoupled resonance frequency f_0 was measured. The measured frequencies were then used to calculate the inductive coupling factor k_L. Using the previously determined capacitance C, the self-inductance L of one coil and the mutual inductance M were calculated.

The measurement procedure for Task 4 was:

1. Assemble the inductively coupled resonant circuits using the two flat coils.
2. Set a distance d between the two coils.
3. Charge the capacitor of one circuit.
4. Start the free oscillation by closing the switch.
5. Record the time trace and FFT spectrum.
6. Determine the two resonance frequencies f_+ and f_-.
7. Repeat the measurement for several coil distances.

For all tasks, the spectra were saved and later plotted with consistent frequency scaling. For Tasks 1 and 2, all ten FFT spectra were plotted together for comparison. For Task 4, the spectra for the different coil distances were also plotted in one graph. The peak frequencies extracted from the spectra formed the basis for all subsequent calculations of coupling factors, capacitance values, beat periods, self-inductance, and mutual inductance.

4 Low-Point Capacitively Coupled Circuit (Task 1)

4.1 Aim of the Measurement

The aim of Task 1 was to investigate the free oscillations of the low-point capacitively coupled circuit for ten different values of the coupling capacitance C_K. For each value of C_K, the frequency spectrum of the measured voltage signal was recorded. From each spectrum, the two normal-mode frequencies were determined. These frequencies were then used to calculate the capacitive coupling factor $k_{C,T}$. Finally, the theoretical dependence of $k_{C,T}$ on C_K was fitted to the measured data in order to determine the capacitance C of the LC circuits.

The coupling capacitances used in this task were

$$C_K = 0.22,\ 0.20,\ 0.18,\ 0.16,\ 0.14,\ 0.12,\ 0.10,\ 0.08,\ 0.06,\ 0.04\ \mu\text{F}. \tag{80}$$

4.2 Relevant Theory

For the low-point capacitively coupled circuit, the two identical LC circuits are coupled by the capacitor C_K. The two normal-mode frequencies are obtained from the frequency splitting in the Fourier spectrum. The experimentally determined coupling factor is calculated from the two measured frequencies using

$$k_{C,T} = \left| \frac{f_1^2 - f_2^2}{f_1^2 + f_2^2} \right|. \tag{81}$$

Here, f_1 and f_2 are the two peak frequencies observed in the spectrum. Since $f_2 > f_1$ in the measurements, this can also be written as

$$k_{C,T} = \frac{f_2^2 - f_1^2}{f_2^2 + f_1^2}. \tag{82}$$

The theoretical expression for the low-point capacitive coupling factor is

$$k_{C,T} = \frac{C}{C + C_K}, \tag{83}$$

where C is the capacitance of each LC circuit and C_K is the coupling capacitance. Equation (83) was used as the fit function. The only fit parameter was C.

4.3 Measurement Method

The low-point circuit was assembled using the two LC resonant circuits and the capacitor decade as the coupling capacitor. For each value of C_K, the capacitors were charged using the switchbox and the free oscillation was started by switching the circuit. The oscilloscope recorded the voltage signal in the time domain and simultaneously produced the corresponding frequency spectrum using the FFT function.

For each of the ten measurements, the frequency spectrum was saved as a CSV file. The files were labelled from `S10.csv` to `S19.csv`, corresponding to the ten coupling capacitances in descending order. The spectra were analysed using Python. The frequency axis of the CSV files was given in kHz and the amplitude was given in dBu.

The FFT frequency spacing was

$$\Delta f = 0.095367 \text{ kHz}. \tag{84}$$

The frequency uncertainty was estimated as half of the FFT bin width:

$$u(f) = \frac{\Delta f}{2} = 0.047684 \text{ kHz}. \tag{85}$$

4.4 Measured Frequency Spectra

Figure 1 shows the ten measured frequency spectra for the low-point circuit. The spectra are plotted on the same amplitude scale without vertical offsets. The two resonance peaks correspond to the two normal modes of the coupled circuit. As the coupling capacitance is decreased, the separation between the two peaks increases. This behaviour is expected from Eq. (83), since the coupling factor $k_{C,T}$ increases when C_K becomes smaller.

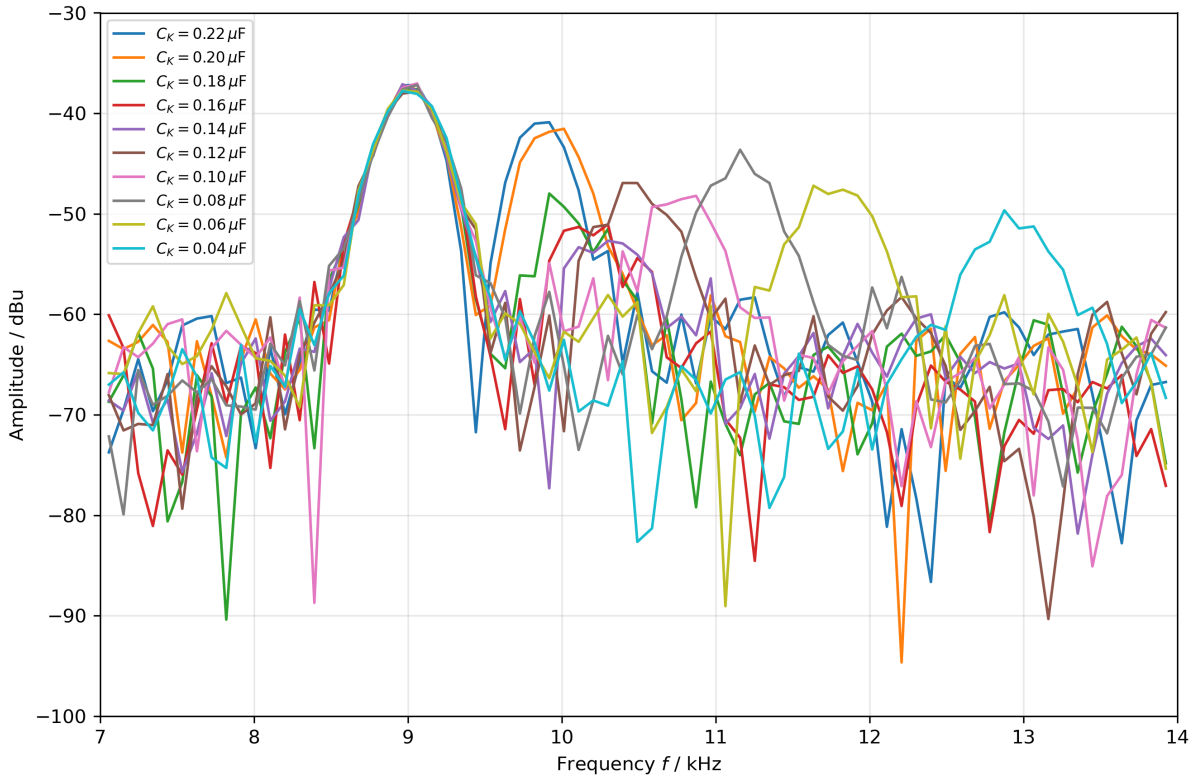


Figure 1: Frequency spectra of the low-point capacitively coupled circuit for ten different values of the coupling capacitance C_K.

The peak positions were identified from the spectra in the frequency interval from 7 to 14 kHz. A second plot with the marked peak positions was used to check the peak selection. This is shown in Fig. 2.

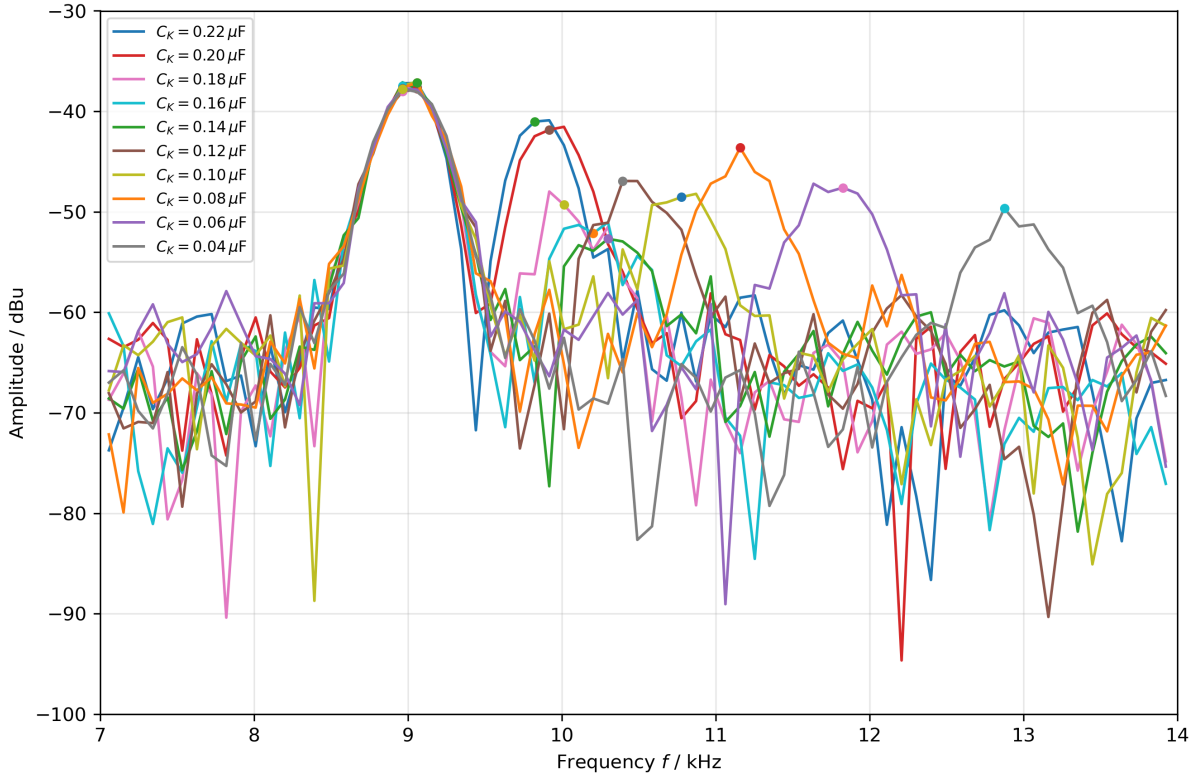


Figure 2: Frequency spectra of the low-point circuit with the selected peak positions used to determine f_1 and f_2.

4.5 Calculation of Coupling Factors

For each measurement, the coupling factor was calculated using Eq. (82). As an example, for the first measurement with $C_K = 0.22\ \mu$F, the measured frequencies were

$$f_1 = 9.0599\ \text{kHz}, \qquad f_2 = 9.8228\ \text{kHz}. \tag{86}$$

The coupling factor is therefore

$$k_{C,T} = \frac{f_2^2 - f_1^2}{f_2^2 + f_1^2} \tag{87}$$

$$= \frac{(9.8228)^2 - (9.0599)^2}{(9.8228)^2 + (9.0599)^2} \tag{88}$$

$$= 0.0807. \tag{89}$$

The measured frequencies and calculated coupling factors are listed in Table 1. The uncertainty of each coupling capacitance was taken as 5%:

$$u(C_K) = 0.05 C_K. \tag{90}$$

Table 1: Measured frequencies and calculated coupling factors for the low-point capacitively coupled circuit.

File	C_K μF	$u(C_K)$ μF	f_1 kHz	$u(f_1)$ kHz	f_2 kHz	$u(f_2)$ kHz	$k_{C,T}$	$u(k_{C,T})$
S10	0.220	0.011	9.0599	0.0477	9.8228	0.0477	0.0807	0.0071
S11	0.200	0.010	9.0599	0.0477	9.9182	0.0477	0.0903	0.0071
S12	0.180	0.009	9.0599	0.0477	10.0136	0.0477	0.0998	0.0070
S13	0.160	0.008	9.0599	0.0477	10.2043	0.0477	0.1184	0.0069
S14	0.140	0.007	9.0599	0.0477	10.2997	0.0477	0.1276	0.0069
S15	0.120	0.006	8.9645	0.0477	10.3951	0.0477	0.1470	0.0069
S16	0.100	0.005	8.9645	0.0477	10.7765	0.0477	0.1820	0.0067
S17	0.080	0.004	9.0599	0.0477	11.1580	0.0477	0.2053	0.0065
S18	0.060	0.003	8.9645	0.0477	11.8256	0.0477	0.2701	0.0062
S19	0.040	0.002	8.9645	0.0477	12.8746	0.0477	0.3470	0.0057

4.6 Uncertainty Analysis

The uncertainty in the frequency determination was estimated from the FFT frequency resolution. Since the frequency spacing was $\Delta f = 0.095367$ kHz, the uncertainty of each peak frequency was taken as

$$u(f_1) = u(f_2) = 0.047684 \text{ kHz}. \tag{91}$$

This estimate assumes that the true peak position lies within approximately half of one FFT bin around the selected maximum.

The uncertainty of the coupling factor was calculated using Gaussian error propagation. With

$$k_{C,T} = \frac{f_2^2 - f_1^2}{f_2^2 + f_1^2}, \tag{92}$$

the uncertainty is

$$u(k_{C,T}) = \sqrt{\left(\frac{\partial k_{C,T}}{\partial f_1} u(f_1)\right)^2 + \left(\frac{\partial k_{C,T}}{\partial f_2} u(f_2)\right)^2}. \tag{93}$$

The required partial derivatives are

$$\frac{\partial k_{C,T}}{\partial f_1} = -\frac{4 f_1 f_2^2}{(f_1^2 + f_2^2)^2}, \tag{94}$$

and

$$\frac{\partial k_{C,T}}{\partial f_2} = \frac{4 f_2 f_1^2}{(f_1^2 + f_2^2)^2}. \tag{95}$$

Therefore,

$$u(k_{C,T}) = \sqrt{\left[-\frac{4 f_1 f_2^2}{(f_1^2 + f_2^2)^2} u(f_1)\right]^2 + \left[\frac{4 f_2 f_1^2}{(f_1^2 + f_2^2)^2} u(f_2)\right]^2}. \tag{96}$$

The uncertainty in the coupling capacitance was taken as 5% of the selected value. This uncertainty was included in the fit as the horizontal uncertainty $u(C_K)$, while $u(k_{C,T})$ was used as the vertical uncertainty.

The main sources of uncertainty in this task are finite frequency resolution of the FFT spectrum, broadening of the resonance peaks due to damping, uncertainty in identifying the exact peak position, tolerance of the coupling capacitor decade, small differences between the two nominally identical LC circuits and parasitic capacitances and resistances in the circuit and wiring.

4.7 Fit of the Theoretical Expression

The measured coupling factors were fitted with the low-point theoretical expression

$$k_{C,T}(C_K) = \frac{C}{C + C_K}. \tag{97}$$

An orthogonal distance regression was used so that both the uncertainty in C_K and the uncertainty in $k_{C,T}$ were included. The fit result was

$$C = 0.022000 \ \mu\text{F} = 22.000 \text{ nF}. \tag{98}$$

The numerical fit uncertainty printed by the analysis rounded to zero at the displayed precision. Therefore, the result is written as

$$C = (22.000 \pm < 0.001) \text{ nF}, \tag{99}$$

where the stated uncertainty represents the numerical precision of the fit output. In practice, the physical uncertainty is expected to be dominated by systematic effects such as component tolerances, peak broadening, and non-identical circuit elements.

The measured coupling factors together with the fitted curve are shown in Fig. 3. The fit follows the expected trend: the coupling factor increases as the coupling capacitance decreases.

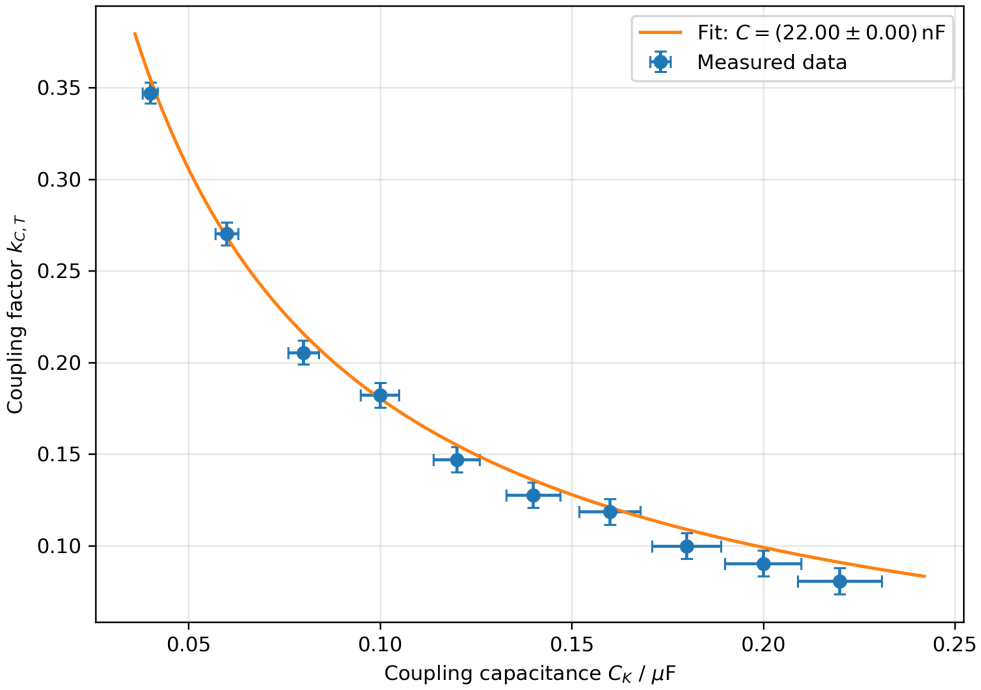


Figure 3: Coupling factor $k_{C,T}$ as a function of coupling capacitance C_K for the low-point circuit.

The fit residuals are shown in Fig. 4. The residuals are mostly negative, with two points close to or slightly above zero. This indicates that the theoretical curve describes the general dependence well, but there may be small systematic deviations. These deviations are likely caused by peak-selection uncertainty, finite FFT resolution, damping, and the fact that the two LC circuits are not perfectly identical.

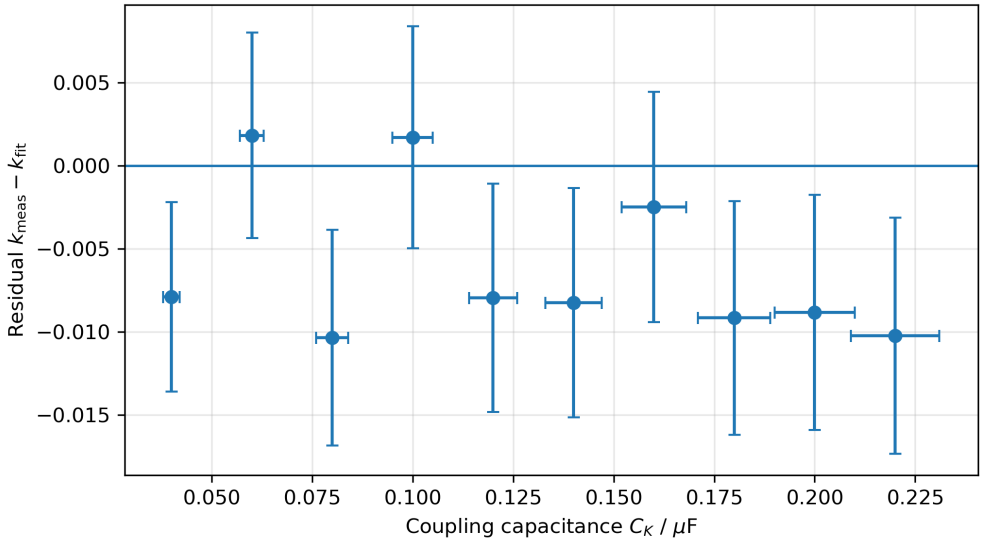


Figure 4: Residuals of the fit for the low-point circuit, calculated as $k_{\text{meas}} - k_{\text{fit}}$.

4.8 Interpretation and Discussion

The measured spectra show two clear resonance peaks for each value of the coupling capacitance. These peaks correspond to the two normal modes of the coupled system. The lower-frequency peak stayed close to 9.0 kHz for all measurements, while the higher-frequency peak shifted from approximately 9.82 kHz to 12.87 kHz as C_K decreased. This increasing frequency splitting shows that the coupling becomes stronger when the coupling capacitance is reduced in the low-point circuit.

The calculated coupling factor increased from

$$k_{C,T} = 0.0807 \tag{100}$$

for $C_K = 0.22$ μF to

$$k_{C,T} = 0.3470 \tag{101}$$

for $C_K = 0.04\ \mu$F. This is consistent with the theoretical expression

$$k_{C,T} = \frac{C}{C + C_K}, \tag{102}$$

which predicts a larger coupling factor for smaller C_K.

The fit gave a capacitance of

$$C = 22.000\ \text{nF}. \tag{103}$$

Since no nominal value of the fixed capacitors was available, a direct comparison with the manufacturer value could not be made. However, the fitted value is physically reasonable for an LC circuit with resonance frequencies near 9 kHz and the available laboratory components. The residuals show deviations of order 10^{-2} in the coupling factor, which are comparable to the expected experimental limitations.

Overall, Task 1 confirms the expected behaviour of the low-point capacitively coupled circuit. The frequency splitting increased with decreasing C_K, the calculated coupling factors followed the theoretical dependence, and the fit allowed the circuit capacitance C to be determined from the measured spectra.

5 High-Point Capacitively Coupled Circuit (Task 2)

5.1 Aim of the Measurement

The aim of Task 2 was to investigate the high-point capacitively coupled circuit for ten different values of the coupling capacitance C_K. For each value of C_K, a frequency spectrum of the free oscillation was recorded. In the ideal high-point configuration, the in-phase and out-of-phase modes can be excited separately by choosing the initial charging conditions $U_a = U_b$ and $U_a = -U_b$, respectively. In the present measurement, both normal modes were visible in each recorded spectrum, so the two peak frequencies were extracted from every spectrum.

The measured normal-mode frequencies were used to calculate the high-point capacitive coupling factor $k_{C,H}$. The theoretical expression for $k_{C,H}$ was then fitted to the data in order to determine the capacitance C of the resonant circuits.

The coupling capacitances used in this task were

$$C_K = 0.01,\ 0.02,\ 0.03,\ 0.05,\ 0.07,\ 0.09,\ 0.10,\ 0.13,\ 0.16,\ 0.20\ \mu\text{F}. \tag{104}$$

5.2 Relevant Theory

For the high-point capacitively coupled circuit, the two normal-mode angular frequencies are

$$\omega_+ = \frac{1}{\sqrt{L(C + 2C_K)}} \tag{105}$$

and

$$\omega_- = \frac{1}{\sqrt{LC}}. \tag{106}$$

Since $C_K > 0$, the lower-frequency mode is ω_+, while the higher-frequency mode is ω_-. Therefore, the corresponding measured frequencies are denoted by f_+ and f_-.

The coupling factor for the high-point circuit is calculated from the measured frequencies using

$$k_{C,H} = \frac{f_-^2 - f_+^2}{f_-^2 + f_+^2}. \tag{107}$$

The theoretical expression for the high-point capacitive coupling factor is

$$k_{C,H} = \frac{C_K}{C + C_K}, \tag{108}$$

where C is the capacitance of each resonant circuit and C_K is the coupling capacitance. Equation (108) was used as the fit function, with C as the only fit parameter.

5.3 Measurement Method

The high-point circuit was assembled using the same two LC resonant circuits as in Task 1. The coupling capacitance C_K was selected using the capacitor decade. For each value of C_K, the capacitors were charged using the switchbox and the resulting free oscillation was measured with the oscilloscope. The oscilloscope recorded the voltage signal and simultaneously generated the corresponding FFT spectrum.

The spectra were saved as CSV files labelled from `S20.csv` to `S29.csv`. The frequency axis in these files was given in MHz, while the amplitude was given in dBu. For the table and calculations, the peak frequencies were converted to kHz.

The FFT frequency spacing was

$$\Delta f = 0.00038147 \text{ MHz} = 0.381470 \text{ kHz}. \tag{109}$$

The uncertainty of each frequency peak was estimated as half of the FFT bin width:

$$u(f) = \frac{\Delta f}{2} = 0.00019073 \text{ MHz} = 0.190735 \text{ kHz}. \tag{110}$$

5.4 Measured Frequency Spectra

Figure 5 shows the ten measured spectra for the high-point circuit. The spectra are plotted on the same amplitude scale without vertical offsets. The frequency range is shown from 0.00 to 0.02 MHz, corresponding to 0 to 20 kHz.

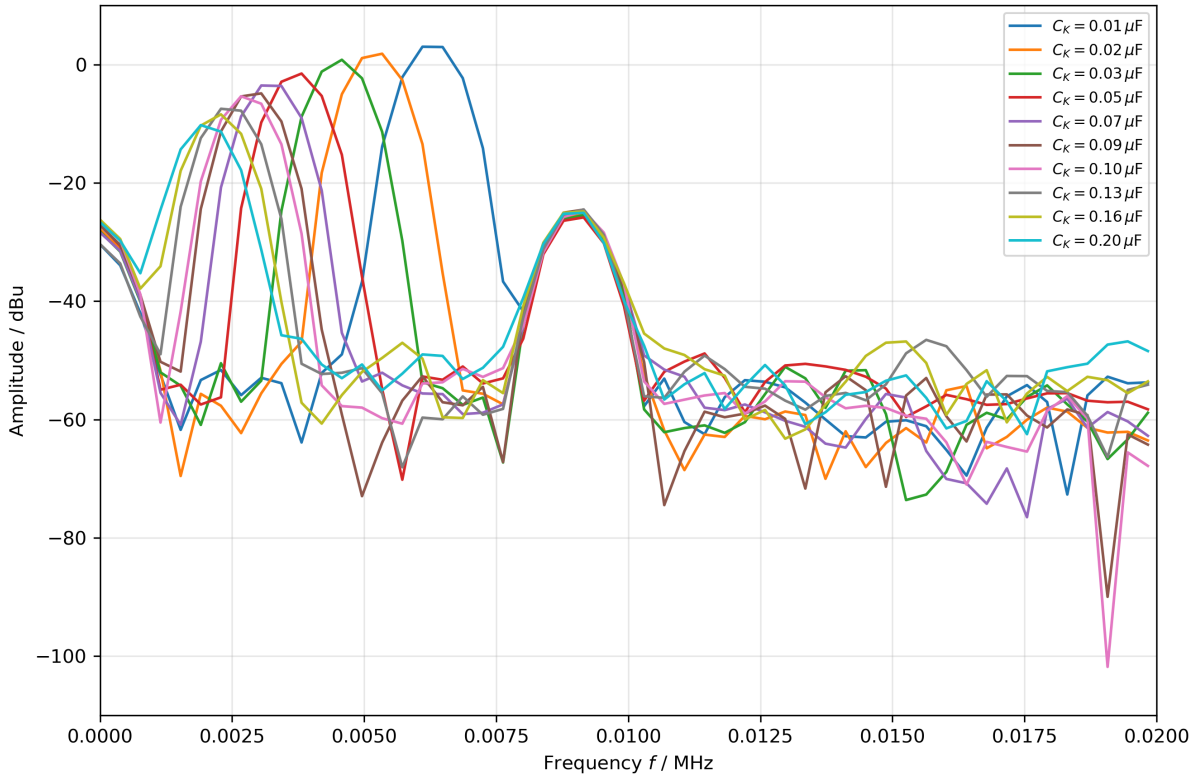


Figure 5: Frequency spectra of the high-point capacitively coupled circuit for ten values of the coupling capacitance C_K.

For each spectrum, the lower-frequency peak was assigned to f_+ and the higher-frequency peak was assigned to f_-. A peak-check plot was produced to verify that the selected peak positions correspond to the visible resonance peaks. This is shown in Fig. 6.

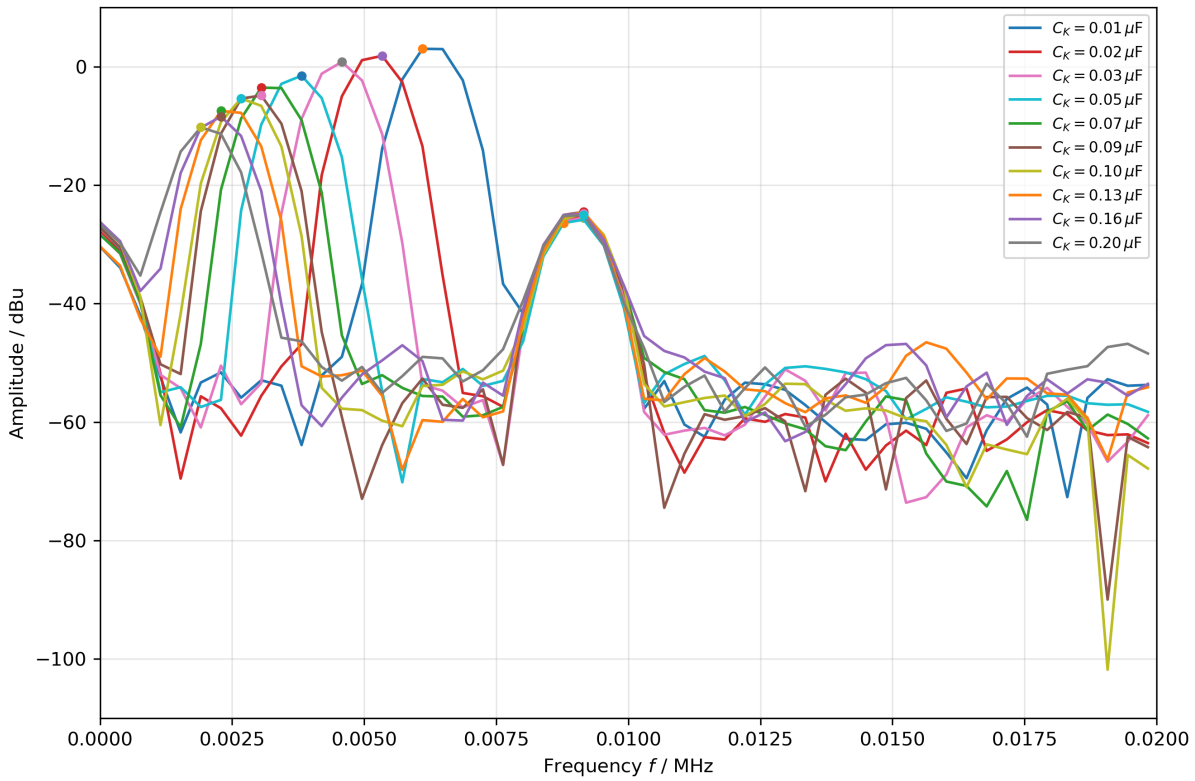


Figure 6: Frequency spectra of the high-point circuit with the selected peak positions used to determine f_+ and f_-.

5.5 Calculation of Coupling Factors

The coupling factor was calculated for each coupling capacitance using Eq. (107). As an example, for the first measurement with $C_K = 0.01\ \mu$F, the measured frequencies were

$$f_+ = 6.1035\ \text{kHz}, \qquad f_- = 9.1553\ \text{kHz}. \tag{111}$$

The coupling factor is therefore

$$k_{C,H} = \frac{f_-^2 - f_+^2}{f_-^2 + f_+^2} \tag{112}$$

$$= \frac{(9.1553)^2 - (6.1035)^2}{(9.1553)^2 + (6.1035)^2} \tag{113}$$

$$= 0.3846. \tag{114}$$

The measured frequencies and calculated coupling factors are listed in Table 2. The uncertainty of the coupling capacitance was taken as 5%:

$$u(C_K) = 0.05 C_K. \tag{115}$$

Table 2: Measured frequencies and calculated coupling factors for the high-point capacitively coupled circuit.

File	C_K μF	$u(C_K)$ μF	f_+ kHz	$u(f_+)$ kHz	f_- kHz	$u(f_-)$ kHz	$k_{C,H}$	$u(k_{C,H})$
S20	0.010	0.0005	6.1035	0.1907	9.1553	0.1907	0.3846	0.0320
S21	0.020	0.0010	5.3406	0.1907	9.1553	0.1907	0.4922	0.0313
S22	0.030	0.0015	4.5776	0.1907	9.1553	0.1907	0.6000	0.0298
S23	0.050	0.0025	3.8147	0.1907	8.7738	0.1907	0.6820	0.0292
S24	0.070	0.0035	3.0518	0.1907	9.1553	0.1907	0.8000	0.0237
S25	0.090	0.0045	3.0518	0.1907	9.1553	0.1907	0.8000	0.0237
S26	0.100	0.0050	2.6703	0.1907	9.1553	0.1907	0.8432	0.0215
S27	0.130	0.0065	2.2888	0.1907	9.1553	0.1907	0.8824	0.0190
S28	0.160	0.0080	2.2888	0.1907	9.1553	0.1907	0.8824	0.0190
S29	0.200	0.0100	1.9074	0.1907	9.1553	0.1907	0.9168	0.0163

5.6 Uncertainty Analysis

The uncertainty in the frequency determination was estimated from the FFT resolution. Since the frequency spacing was

$$\Delta f = 0.381470 \text{ kHz}, \tag{116}$$

the uncertainty of each peak frequency was taken as

$$u(f_+) = u(f_-) = 0.190735 \text{ kHz}. \tag{117}$$

This is a conservative estimate based on the assumption that the true peak position lies within approximately half of one FFT bin around the selected maximum.

The uncertainty of the coupling factor was calculated using Gaussian error propagation. With

$$k_{C,H} = \frac{f_-^2 - f_+^2}{f_-^2 + f_+^2}, \tag{118}$$

the propagated uncertainty is

$$u(k_{C,H}) = \sqrt{\left(\frac{\partial k_{C,H}}{\partial f_+} u(f_+)\right)^2 + \left(\frac{\partial k_{C,H}}{\partial f_-} u(f_-)\right)^2}. \tag{119}$$

The partial derivatives are

$$\frac{\partial k_{C,H}}{\partial f_+} = -\frac{4 f_+ f_-^2}{(f_+^2 + f_-^2)^2}, \tag{120}$$

and

$$\frac{\partial k_{C,H}}{\partial f_-} = \frac{4 f_- f_+^2}{(f_+^2 + f_-^2)^2}. \tag{121}$$

Thus,

$$u(k_{C,H}) = \sqrt{\left[-\frac{4 f_+ f_-^2}{(f_+^2 + f_-^2)^2} u(f_+)\right]^2 + \left[\frac{4 f_- f_+^2}{(f_+^2 + f_-^2)^2} u(f_-)\right]^2}. \tag{122}$$

The coupling capacitor uncertainty was taken as

$$u(C_K) = 0.05 C_K. \tag{123}$$

Both the horizontal uncertainty $u(C_K)$ and the vertical uncertainty $u(k_{C,H})$ were included in the fit by using orthogonal distance regression.

The main uncertainty sources in this task were finite FFT frequency resolution, broadening of the resonance peaks due to damping, uncertainty in assigning the exact peak positions, tolerance of the capacitor decade, possible overlap or weak visibility of the low-frequency peak for large coupling capacitances, deviations from the assumption of two identical LC circuits and parasitic capacitances and resistances in the wiring and components.

5.7 Fit of the Theoretical Expression

The measured coupling factors were fitted using

$$k_{C,H}(C_K) = \frac{C_K}{C + C_K}. \tag{124}$$

The only fit parameter was the capacitance C of the resonant circuits. The fit gave

$$C = 0.022000 \ \mu\text{F} = 22.000 \text{ nF}. \tag{125}$$

The numerical fit uncertainty printed by the analysis rounded to zero at the displayed precision. Therefore, the fit result is reported as

$$C = (22.000 \pm < 0.001) \text{ nF}. \tag{126}$$

This very small numerical uncertainty should not be interpreted as the full experimental uncertainty. The physical uncertainty is mainly determined by systematic effects such as FFT peak selection, component tolerances, damping, and non-ideal circuit symmetry.

The measured data and fitted curve are shown in Fig. 7. The coupling factor increases with increasing C_K, as expected from the high-point expression.

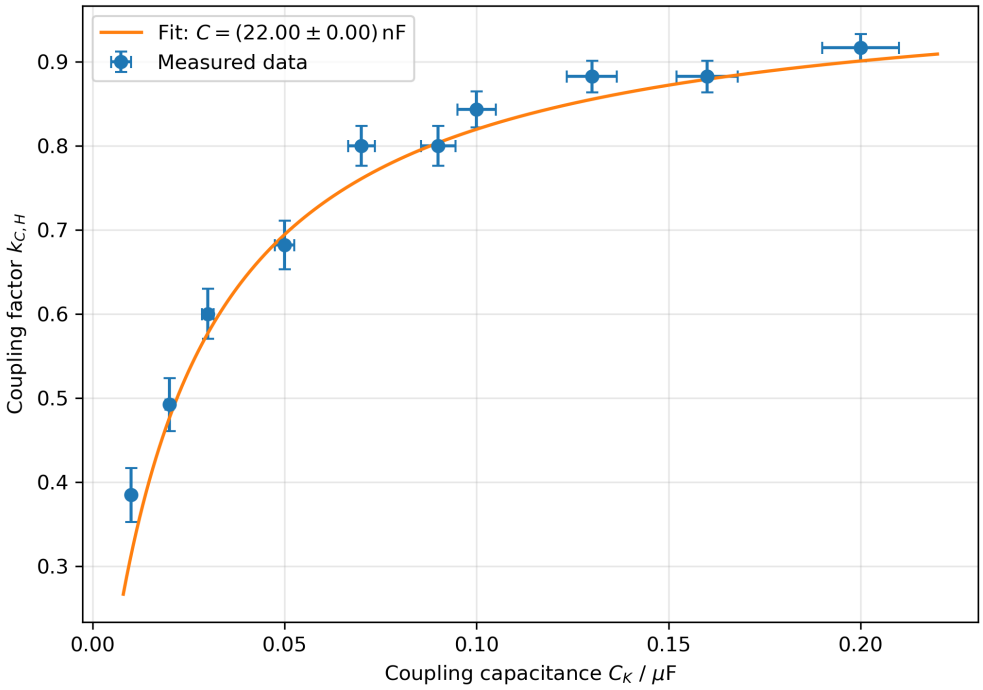


Figure 7: Coupling factor $k_{C,H}$ as a function of coupling capacitance C_K for the high-point circuit. The curve shows the fit using $k_{C,H} = C_K/(C + C_K)$.

The fit residuals are shown in Fig. 8. Most residuals are positive, especially for small coupling capacitances. This means that several measured coupling factors are slightly larger than the fitted theoretical values. The largest deviation occurs at $C_K = 0.01$ μF, where the measured coupling factor is 0.3846 while the fitted value is 0.3125. This point is also more sensitive to the uncertainty in the lower-frequency peak, because the high-point coupling factor depends strongly on the frequency separation.

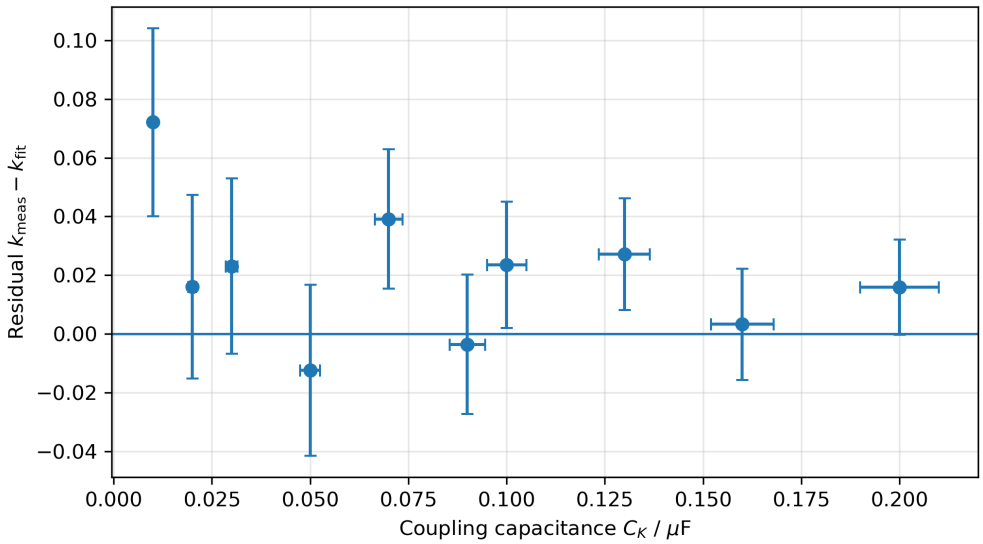


Figure 8: Residuals of the fit for the high-point circuit, calculated as $k_{\text{meas}} - k_{\text{fit}}$.

5.8 Interpretation and Discussion

The high-point circuit showed two resonance peaks in each measured spectrum. The higher-frequency peak remained approximately constant near 9.16 kHz for most measurements, while the lower-frequency peak shifted to smaller frequencies as C_K increased. This behaviour agrees with the theoretical normal-mode frequencies

$$\omega_+ = \frac{1}{\sqrt{L(C + 2C_K)}}, \qquad \omega_- = \frac{1}{\sqrt{LC}}. \tag{127}$$

The frequency ω_- is independent of C_K, while ω_+ decreases as the coupling capacitance increases.

The calculated coupling factor increased from

$$k_{C,H} = 0.3846 \tag{128}$$

for $C_K = 0.01$ μF to

$$k_{C,H} = 0.9168 \tag{129}$$

for $C_K = 0.20$ μF. This is consistent with the theoretical expression

$$k_{C,H} = \frac{C_K}{C + C_K}, \tag{130}$$

which predicts stronger coupling for larger values of C_K.

The fitted capacitance was

$$C = 22.000 \text{ nF}. \tag{131}$$

This agrees with the value obtained from the low-point circuit in Task 1. The agreement between the two independently analysed circuit configurations supports the consistency of the capacitive-coupling model. Since no nominal value of the fixed capacitors was available, the fitted values from Tasks 1 and 2 provide the best estimate of the capacitance used in the experiment.

Some deviations from the fit are visible in the residual plot. These deviations are expected because the analysis assumes identical LC circuits and negligible damping, while the real circuit contains resistance, parasitic capacitance, and small component differences. In addition, the frequency resolution in Task 2 was coarser than in Task 1, with $u(f) = 0.190735$ kHz. This affects the precision of the extracted peak positions, especially for low-frequency peaks.

Overall, Task 2 confirms the expected behaviour of the high-point capacitively coupled circuit. The measured frequency spectra show the normal-mode splitting, the calculated coupling factor increases with C_K, and the fitted capacitance is consistent with the value obtained from the low-point circuit.

6 Beat Oscillation in the High-Point Circuit (Task 3)

6.1 Aim of the Measurement

The aim of Task 3 was to measure a beat oscillation in the high-point capacitively coupled circuit for one selected coupling capacitance. In this measurement, one resonant circuit was initially charged, while the other circuit was connected to zero potential. This initial condition excites both normal modes simultaneously. The superposition of the two close normal-mode frequencies produces a fast oscillation whose amplitude is modulated periodically, resulting in a beat signal.

The selected coupling capacitance was

$$C_K = 0.004 \ \mu\text{F}. \tag{132}$$

The uncertainty of the coupling capacitance was again taken as 5%:

$$u(C_K) = 0.05 C_K = 0.0002 \ \mu\text{F}. \tag{133}$$

For this task, both the time trace and the corresponding frequency spectrum were recorded. The main objective was to determine the beat period T_S directly from the time trace and compare it with the value calculated from the two normal-mode frequencies observed in the spectrum. This directly checks the relation between the time-domain beat signal and the frequency-domain mode splitting.

6.2 Relevant Theory

A beat oscillation occurs when two oscillations with similar frequencies are superimposed. If the two normal modes have angular frequencies ω_+ and ω_-, the measured signal may be approximated as

$$U(t) = U_0 \cos(\omega_+ t) + U_0 \cos(\omega_- t). \tag{134}$$

Using the identity

$$\cos a + \cos b = 2 \cos\left(\frac{a-b}{2}\right) \cos\left(\frac{a+b}{2}\right), \tag{135}$$

this becomes

$$U(t) = 2U_0 \cos\left(\frac{\omega_- - \omega_+}{2} t\right) \cos\left(\frac{\omega_- + \omega_+}{2} t\right). \tag{136}$$

The second cosine factor describes the fast oscillation with an average angular frequency

$$\bar{\omega} = \frac{\omega_- + \omega_+}{2}. \tag{137}$$

Therefore, the corresponding oscillation period is

$$T = \frac{2\pi}{\bar{\omega}} = \frac{4\pi}{\omega_- + \omega_+}. \tag{138}$$

Since $\omega = 2\pi f$, this can be written as

$$T = \frac{2}{f_+ + f_-}. \tag{139}$$

If the frequencies are expressed in kHz, the period obtained from Eq. (139) is in ms.

The slowly varying envelope is determined by the frequency difference. The beat angular frequency is

$$\omega_S = |\omega_- - \omega_+|. \tag{140}$$

Thus, the beat period is

$$T_S = \frac{2\pi}{|\omega_- - \omega_+|} = \frac{1}{|f_- - f_+|}. \tag{141}$$

The high-point coupling factor can also be calculated from the two measured frequencies:

$$k_{C,H} = \frac{f_-^2 - f_+^2}{f_-^2 + f_+^2}. \tag{142}$$

6.3 Measurement Method

The high-point circuit from Task 2 was used. One circuit was connected to zero potential, while the other circuit was charged using the switchbox. After switching, the charged capacitor discharged through the coupled system and a beat oscillation was observed in the measured voltage signal.

Two CSV files were recorded:

- `T3.csv`: time trace of the voltage signal,
- `S3.csv`: FFT spectrum of the same beat oscillation.

The time trace contained the voltage in volts as a function of time in ms. The frequency spectrum contained the amplitude in dBu as a function of frequency in MHz. For the numerical analysis, the spectral peak frequencies were converted to kHz.

The frequency spacing in the spectrum was

$$\Delta f = 0.00038147 \text{ MHz} = 0.381470 \text{ kHz}. \tag{143}$$

The frequency uncertainty was estimated as half of the FFT bin width:

$$u(f) = \frac{\Delta f}{2} = 0.00019073 \text{ MHz} = 0.190735 \text{ kHz}. \tag{144}$$

The time spacing of the time trace was

$$\Delta t = 0.00032000 \text{ ms} = 0.3200 \ \mu\text{s}. \tag{145}$$

The time uncertainty for direct cursor-like time measurements was estimated as

$$u(t) = \frac{\Delta t}{2} = 0.00016000 \text{ ms} = 0.1600 \ \mu\text{s}. \tag{146}$$

6.4 Time Trace and Frequency Spectrum

The measured time trace is shown in Fig. 9. The signal shows the characteristic behaviour of a beat oscillation: a fast oscillation with a slowly varying amplitude envelope. The envelope was determined numerically from the time signal and used to estimate the beat period.

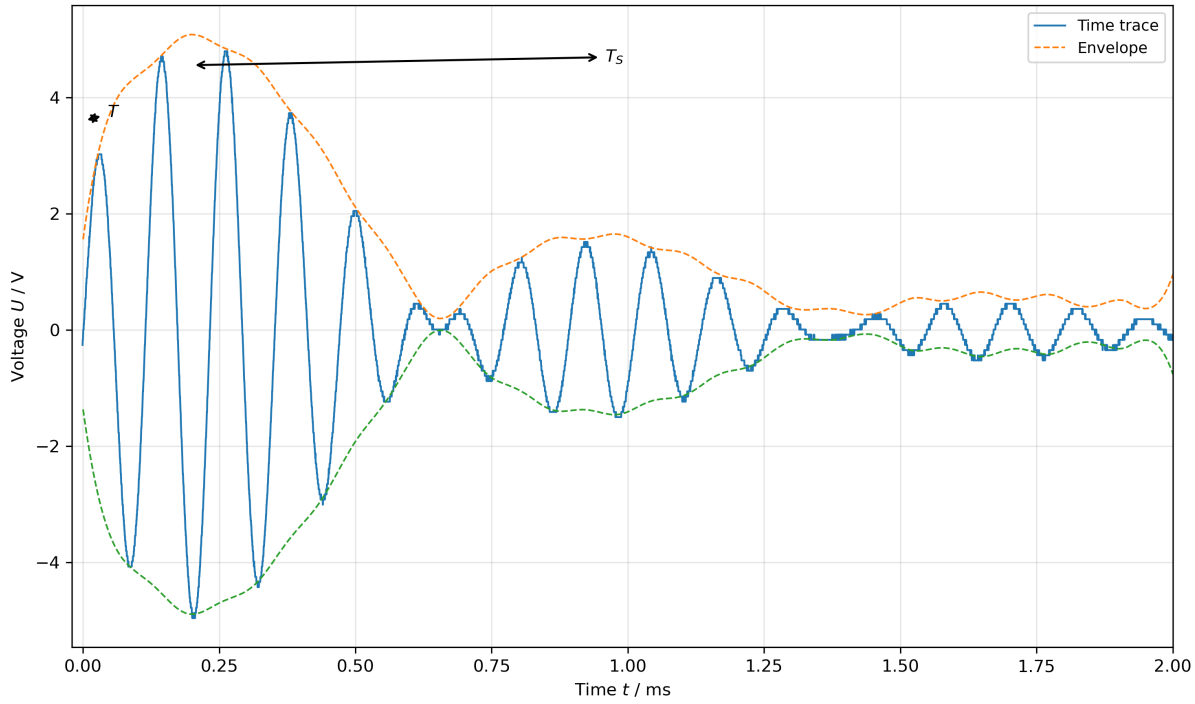


Figure 9: Measured time trace of the beat oscillation in the high-point circuit for $C_K = 0.004$ μF.

The corresponding frequency spectrum is shown in Fig. 10. Two clear peaks are visible. These peaks correspond to the two normal modes that form the beat oscillation in the time domain.

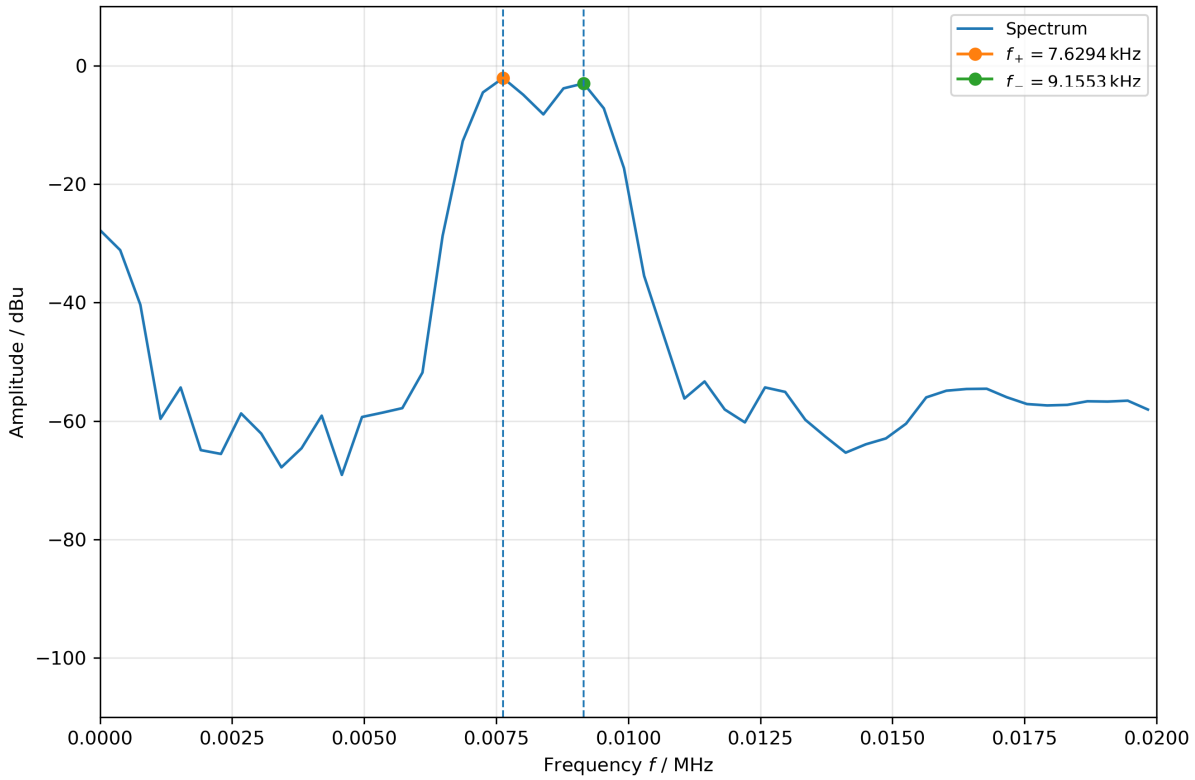


Figure 10: Frequency spectrum of the beat oscillation. The two marked peaks correspond to the normal-mode frequencies f_+ and f_-.

An additional envelope-check plot was used to verify the automatic envelope analysis. This is shown in Fig. 11. The detected envelope extrema were used to estimate the beat period directly from the time trace.

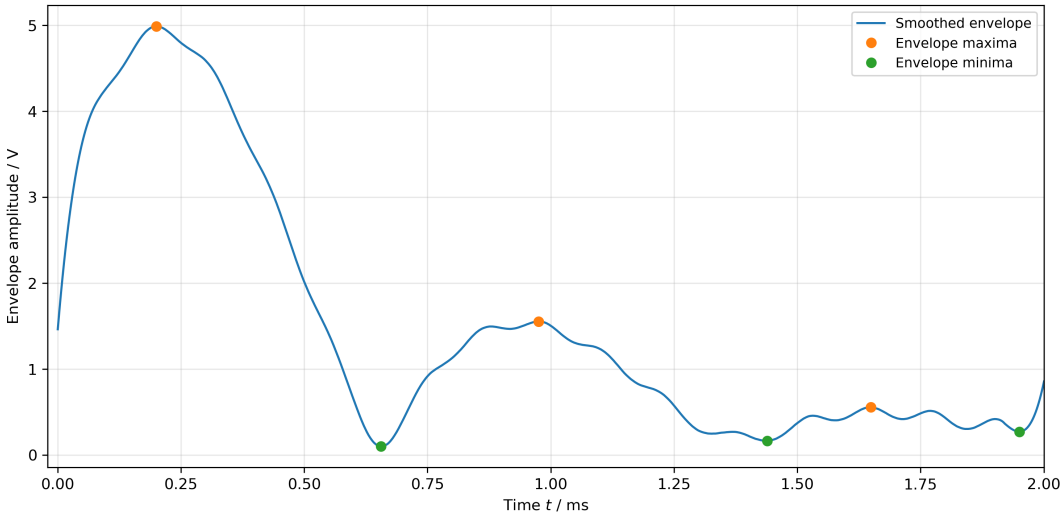


Figure 11: Smoothed amplitude envelope used for determining the beat period from the time trace.

6.5 Frequency-Domain Results

From the frequency spectrum, the two normal-mode frequencies were determined as

$$f_+ = (7.6294 \pm 0.1907)\ \text{kHz}, \tag{147}$$

and

$$f_- = (9.1553 \pm 0.1907)\ \text{kHz}. \tag{148}$$

Using Eq. (142), the coupling factor was

$$k_{C,H} = 0.1803 \pm 0.0315. \tag{149}$$

The frequency-domain results are summarized in Table 3.

Table 3: Frequency-domain results for the beat oscillation measurement.

Quantity	Value	Unit
C_K	0.0040	μF
$u(C_K)$	0.0002	μF
f_+	7.6294	kHz
$u(f_+)$	0.1907	kHz
f_-	9.1553	kHz
$u(f_-)$	0.1907	kHz
$k_{C,H}$	0.1803	–
$u(k_{C,H})$	0.0315	–

Using Eqs. (139) and (141), the periods calculated from the spectral frequencies were

$$T_{\text{calc}} = \frac{2}{f_+ + f_-} = 0.11916\ \text{ms}, \tag{150}$$

and

$$T_{S,\text{calc}} = \frac{1}{f_- - f_+} = 0.65536\ \text{ms}. \tag{151}$$

6.6 Time-Domain Results

The beat period was determined directly from the time trace using the extracted amplitude envelope. The detected beat intervals were

$$0.77504,\ 0.67392,\ 0.78368,\ 0.51104\ \text{ms}. \tag{152}$$

Averaging these intervals gave

$$T_{S,\text{direct}} = (0.68592 \pm 0.06339)\ \text{ms}. \tag{153}$$

The fast oscillation period was also estimated automatically from the time trace using zero crossings. However, this automatic extraction produced

$$T_{\text{direct}} = (0.00064 \pm 0.00045)\ \text{ms}. \tag{154}$$

This value is not physically consistent with the measured resonance frequencies near 8–9 kHz, which imply a fast oscillation period of approximately 0.12 ms. Therefore, the automatically extracted value of T_{direct} is treated as unreliable. The likely reason is that the zero-crossing algorithm selected small numerical or noise-related crossings instead of full oscillation periods. In contrast, the beat period extracted from the envelope is physically meaningful and is the main quantity required for the comparison in this task.

6.7 Uncertainty Analysis

The uncertainty of the coupling factor was calculated by Gaussian error propagation from Eq. (142). With

$$k_{C,H} = \frac{f_-^2 - f_+^2}{f_-^2 + f_+^2}, \tag{155}$$

the uncertainty is

$$u(k_{C,H}) = \sqrt{\left(\frac{\partial k_{C,H}}{\partial f_+} u(f_+)\right)^2 + \left(\frac{\partial k_{C,H}}{\partial f_-} u(f_-)\right)^2}. \tag{156}$$

The partial derivatives are

$$\frac{\partial k_{C,H}}{\partial f_+} = -\frac{4 f_+ f_-^2}{(f_+^2 + f_-^2)^2}, \tag{157}$$

and

$$\frac{\partial k_{C,H}}{\partial f_-} = \frac{4 f_- f_+^2}{(f_+^2 + f_-^2)^2}. \tag{158}$$

The uncertainty of the calculated fast oscillation period was obtained from

$$T_{\mathrm{calc}} = \frac{2}{f_+ + f_-}. \tag{159}$$

Therefore,

$$u(T_{\mathrm{calc}}) = \frac{2\sqrt{u(f_+)^2 + u(f_-)^2}}{(f_+ + f_-)^2}. \tag{160}$$

For the calculated beat period,

$$T_{S,\mathrm{calc}} = \frac{1}{f_- - f_+}, \tag{161}$$

the uncertainty is

$$u(T_{S,\mathrm{calc}}) = \frac{\sqrt{u(f_+)^2 + u(f_-)^2}}{(f_- - f_+)^2}. \tag{162}$$

The uncertainty of the directly measured beat period was estimated from the spread of the detected envelope intervals and the time resolution of the oscilloscope. The main uncertainty sources in this task are finite FFT frequency resolution, uncertainty in identifying the two spectral peak positions, damping and broadening of the resonance peaks, noise in the time trace, uncertainty in extracting the beat envelope, finite time resolution of the oscilloscope and possible systematic error caused by automatic zero-crossing detection.

6.8 Comparison of Direct and Frequency-Domain Periods

The comparison between the directly measured periods and the periods calculated from the frequency spectrum is shown in Table 4.

Table 4: Comparison of direct time-domain periods with values calculated from the frequency spectrum.

Quantity	Direct ms	u(Direct) ms	Calculated ms	u(Calculated) ms	Relative difference %	z
T	0.00064	0.00045	0.11916	0.00191	99.46	60.23
T_S	0.68592	0.06339	0.65536	0.11585	4.66	0.23

For the beat period, the directly measured value and the value calculated from the spectral splitting agree within uncertainty:

$$T_{S,\mathrm{direct}} = (0.68592 \pm 0.06339)\ \mathrm{ms}, \tag{163}$$

while

$$T_{S,\mathrm{calc}} = (0.65536 \pm 0.11585)\ \mathrm{ms}. \tag{164}$$

The relative difference is only 4.66%, and the normalized deviation is

$$z = 0.23. \tag{165}$$

Since this is much smaller than 1, the two beat-period values are compatible within the estimated uncertainties.

The comparison for the fast oscillation period gives a very large discrepancy. The calculated value,

$$T_{\text{calc}} = (0.11916 \pm 0.00191)\ \text{ms}, \tag{166}$$

is consistent with oscillations at approximately 8–9 kHz. The automatically extracted direct value,

$$T_{\text{direct}} = (0.00064 \pm 0.00045)\ \text{ms}, \tag{167}$$

would correspond to a frequency in the MHz range and is therefore not compatible with the observed spectrum. This indicates that the automatic zero-crossing method failed for the fast oscillation period. For the physical interpretation, the frequency-domain value of T is therefore more reliable.

6.9 Interpretation and Discussion

The measured time trace clearly shows a beat oscillation, confirming that both normal modes of the coupled circuit were excited at the same time. The frequency spectrum shows two normal-mode peaks at approximately

$$f_+ = 7.6294\ \text{kHz} \tag{168}$$

and

$$f_- = 9.1553\ \text{kHz}. \tag{169}$$

The difference between these two frequencies determines the beat period. Using the spectral splitting gives

$$T_{S,\text{calc}} = 0.65536\ \text{ms}. \tag{170}$$

This agrees well with the envelope-based time-domain result

$$T_{S,\text{direct}} = 0.68592\ \text{ms}. \tag{171}$$

The agreement between these two values verifies the expected relation

$$T_S = \frac{1}{|f_- - f_+|}. \tag{172}$$

This confirms the connection between the frequency-domain splitting of the two normal modes and the amplitude modulation observed in the time-domain beat signal.

The calculated coupling factor for this measurement was

$$k_{C,H} = 0.1803 \pm 0.0315. \tag{173}$$

This value is reasonable for a relatively small coupling capacitance. However, the stated value $C_K = 0.004\ \mu\text{F}$ is smaller than the coupling capacitances used in Task 2. If the selected value were actually $0.04\ \mu\text{F}$ instead of $0.004\ \mu\text{F}$, the expected coupling factor from the fitted high-point relation would be closer to the measured value. Therefore, the capacitance setting should be checked if the original laboratory notes or photographs of the capacitor decade are available.

The main successful result of Task 3 is the agreement of the measured beat period with the value calculated from the frequency splitting. The failed automatic extraction of the fast oscillation period shows that direct time-domain period measurements must be checked carefully, especially when using automated zero-crossing algorithms on noisy or damped signals. A manual cursor measurement over several visible oscillation periods would provide a more reliable direct value for T.

7 Inductively Coupled Resonant Circuits (Task 4)

7.1 Aim of the Measurement

The aim of Task 4 was to investigate two inductively coupled resonant circuits as a function of the distance between the two coils. In this part of the experiment, the coupling was not produced by a coupling capacitor, but by the magnetic field of one coil inducing a voltage in the second coil. The strength of this interaction is described by the mutual inductance M and by the inductive coupling factor k_L.

Frequency spectra were recorded for eight different coil distances:

$$d = 4,\ 6,\ 8,\ 10,\ 12,\ 14,\ 16,\ 18\ \text{cm}. \tag{174}$$

The uncertainty of the distance measurement was estimated as

$$u(d) = 0.1 \text{ cm}. \tag{175}$$

For each distance, the two split resonance frequencies f_+ and f_- were determined from the measured frequency spectrum. These frequencies were used to calculate the coupling factor k_L. The mutual inductance M was then estimated using the capacitance determined from the previous tasks,

$$C = 22.0 \text{ nF}. \tag{176}$$

Since no separate uncoupled resonance frequency f_0 was measured, f_0 was estimated from the measured split frequencies. This is an important limitation of the analysis and is discussed below.

7.2 Relevant Theory

For two identical inductively coupled LC circuits, the normal-mode angular frequencies are

$$\omega_+ = \frac{1}{\sqrt{C(L+M)}}, \tag{177}$$

and

$$\omega_- = \frac{1}{\sqrt{C(L-M)}}. \tag{178}$$

Here, L is the self-inductance of one coil and M is the mutual inductance between the two coils. The inductive coupling factor is defined as

$$k_L = \frac{M}{L}. \tag{179}$$

From the two measured split frequencies, the coupling factor is calculated using

$$k_L = \frac{f_-^2 - f_+^2}{f_-^2 + f_+^2}. \tag{180}$$

The uncoupled resonance frequency is related to L and C by

$$f_0 = \frac{1}{2\pi\sqrt{LC}}. \tag{181}$$

Therefore, if f_0 is known, the self-inductance can be calculated as

$$L = \frac{1}{(2\pi f_0)^2 C}. \tag{182}$$

The mutual inductance is then

$$M = k_L L. \tag{183}$$

In this experiment, f_0 was not measured independently. Therefore, it was estimated from the two split frequencies. Starting from the inductive-coupling relations, one obtains

$$f_0^2 = \frac{2 f_+^2 f_-^2}{f_+^2 + f_-^2}. \tag{184}$$

This estimate was used only because no separate uncoupled measurement was available.

The distance dependence of the mutual inductance was analysed using a power law,

$$M(d) = A d^{-n}, \tag{185}$$

where A is a proportionality constant and n is the power-law exponent. Taking the logarithm gives

$$\ln M = \ln A - n \ln d. \tag{186}$$

Thus, the exponent n can be obtained from the slope of a log-log plot.

For a circular coil, the magnetic field on the symmetry axis is approximately proportional to d^{-3} only in the far-field limit. Therefore, a value close to $n = 3$ would be expected only if the distance between the coils is much larger than the coil radius and if the coils are well aligned.

7.3 Measurement Method

The two resonant circuits were coupled inductively by placing two flat circular coils opposite each other on an optical bench. The distance d between the coils was varied from 4 cm to 18 cm. For each distance, the capacitor in one circuit was charged using the switchbox and the resulting free oscillation was recorded with the oscilloscope. The FFT spectrum was saved for each measurement.

The spectra were saved as CSV files labelled from `S41.csv` to `S48.csv`. The file-to-distance assignment was:

File	S41	S42	S43	S44	S45	S46	S47	S48
d/cm	4	6	8	10	12	14	16	18

(187)

The frequency axis in the spectra was given in MHz and was converted to kHz for the calculations. The FFT frequency spacing was

$$\Delta f = 0.00038147 \text{ MHz} = 0.381470 \text{ kHz}. \tag{188}$$

The frequency uncertainty was estimated as half of the FFT bin width:

$$u(f) = \frac{\Delta f}{2} = 0.00019073 \text{ MHz} = 0.190735 \text{ kHz}. \tag{189}$$

7.4 Measured Frequency Spectra

Figure 12 shows the measured frequency spectra for all eight coil distances. The frequency splitting is largest at the smallest distance and becomes smaller as the distance increases. This is expected because the magnetic coupling between the coils becomes weaker when the coils are moved farther apart.

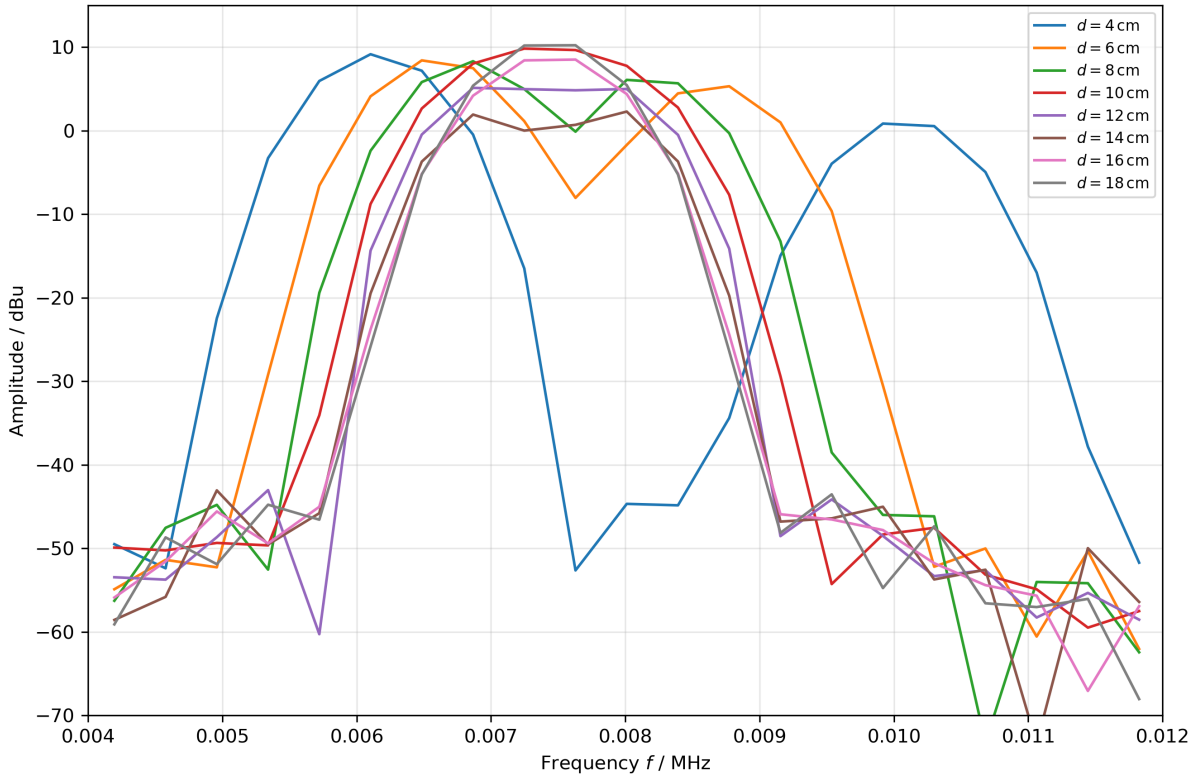


Figure 12: Frequency spectra of the inductively coupled resonant circuits for different coil distances.

The selected peak positions are shown in Fig. 13. The lower-frequency peak was assigned to f_+ and the higher-frequency peak to f_-. At larger distances, the peaks are closer together, so the uncertainty in the peak positions has a larger influence on the calculated coupling factor.

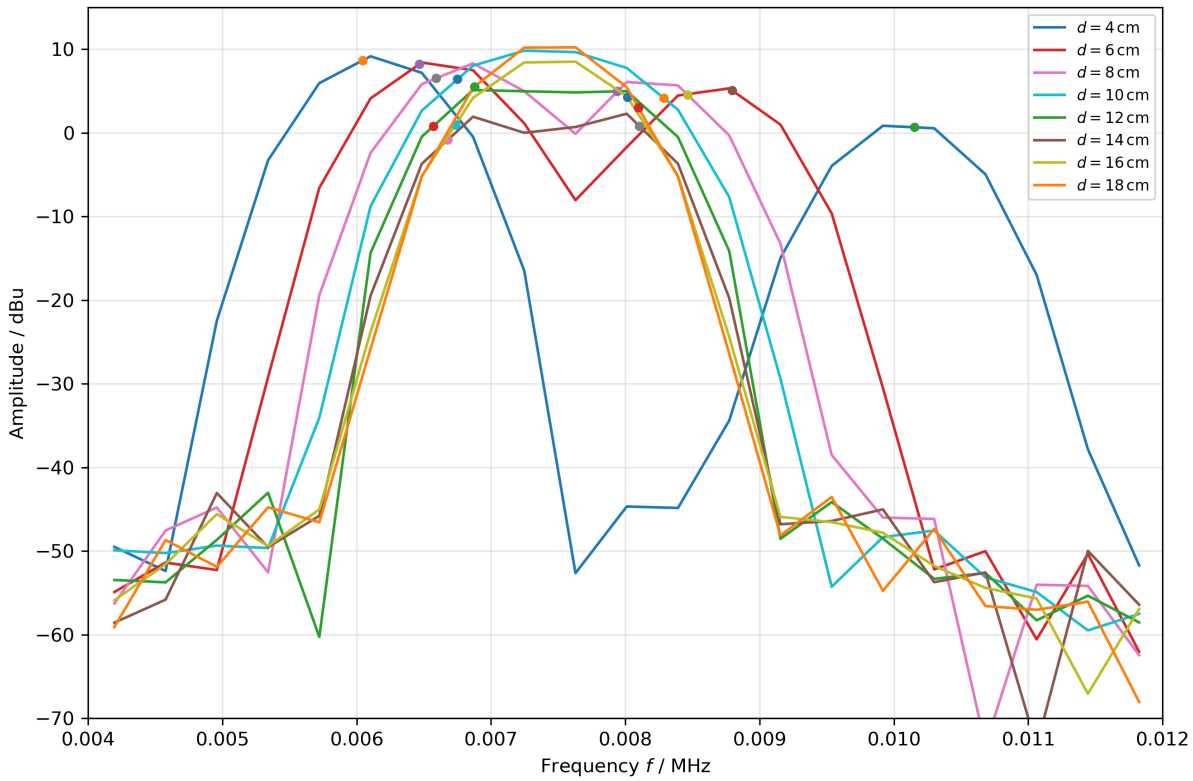


Figure 13: Frequency spectra with the selected peak positions used to determine f_+ and f_-.

7.5 Calculation of Coupling Factor, Self-Inductance, and Mutual Inductance

For each distance, the inductive coupling factor was calculated from Eq. (180). As an example, for $d = 4$ cm, the measured frequencies were

$$f_+ = 6.0465 \text{ kHz}, \qquad f_- = 10.1519 \text{ kHz}. \tag{190}$$

Thus,

$$k_L = \frac{f_-^2 - f_+^2}{f_-^2 + f_+^2} \tag{191}$$

$$= \frac{(10.1519)^2 - (6.0465)^2}{(10.1519)^2 + (6.0465)^2} \tag{192}$$

$$= 0.4763. \tag{193}$$

Since the uncoupled resonance frequency was not measured separately, f_0 was estimated from Eq. (184). The weighted mean of the estimated uncoupled frequencies was

$$f_{0,\text{est}} = (7.3266 \pm 0.0518) \text{ kHz}. \tag{194}$$

Using $C = 22.0$ nF, the self-inductance was calculated as

$$L = (21.4493 \pm 0.3034) \text{ mH}. \tag{195}$$

The mutual inductance was then calculated for each distance using

$$M = k_L L. \tag{196}$$

The full set of results is shown in Table 5.

Table 5: Measured split frequencies, inductive coupling factors, estimated uncoupled frequencies, and mutual inductances for the inductively coupled resonant circuits.

File	d cm	$u(d)$ cm	f_+ kHz	$u(f_+)$ kHz	f_- kHz	$u(f_-)$ kHz	k_L	$u(k_L)$	$f_{0,\mathrm{est}}$ kHz	$u(f_{0,\mathrm{est}})$ kHz	M mH	$u(M)$ mH
S41	4.0	0.1	6.0465	0.1907	10.1519	0.1907	0.4763	0.0284	7.3466	0.1743	10.2164	0.6258
S42	6.0	0.1	6.4677	0.1907	8.7949	0.1907	0.2980	0.0334	7.3687	0.1513	6.3924	0.7211
S43	8.0	0.1	6.5922	0.1907	8.4643	0.1907	0.2449	0.0345	7.3553	0.1466	5.2526	0.7432
S44	10.0	0.1	6.7513	0.1907	8.2876	0.1907	0.2022	0.0349	7.4024	0.1434	4.3373	0.7521
S45	12.0	0.1	6.5719	0.1907	7.9385	0.1907	0.1867	0.0364	7.1591	0.1415	4.0048	0.7820
S46	14.0	0.1	6.6783	0.1907	8.1042	0.1907	0.1911	0.0357	7.2887	0.1420	4.0996	0.7670
S47	16.0	0.1	6.7362	0.1907	8.0158	0.1907	0.1722	0.0359	7.2931	0.1412	3.6933	0.7716
S48	18.0	0.1	6.8762	0.1907	8.0981	0.1907	0.1621	0.0354	7.4126	0.1399	3.4774	0.7616

7.6 Uncertainty Analysis

The frequency uncertainty was estimated from the FFT bin width:

$$u(f_+) = u(f_-) = 0.190735 \text{ kHz}. \tag{197}$$

The distance uncertainty was

$$u(d) = 0.1 \text{ cm}. \tag{198}$$

The uncertainty of the coupling factor was obtained by Gaussian error propagation. With

$$k_L = \frac{f_-^2 - f_+^2}{f_-^2 + f_+^2}, \tag{199}$$

the uncertainty is

$$u(k_L) = \sqrt{\left(\frac{\partial k_L}{\partial f_+} u(f_+)\right)^2 + \left(\frac{\partial k_L}{\partial f_-} u(f_-)\right)^2}. \tag{200}$$

The partial derivatives are

$$\frac{\partial k_L}{\partial f_+} = -\frac{4 f_+ f_-^2}{(f_+^2 + f_-^2)^2}, \tag{201}$$

and

$$\frac{\partial k_L}{\partial f_-} = \frac{4 f_- f_+^2}{(f_+^2 + f_-^2)^2}. \tag{202}$$

The uncertainty of the estimated uncoupled resonance frequency was obtained numerically by propagating the uncertainties of f_+ and f_- through Eq. (184). The final average value was

$$f_{0,\mathrm{est}} = (7.3266 \pm 0.0518) \text{ kHz}. \tag{203}$$

For the self-inductance,

$$L = \frac{1}{(2\pi f_0)^2 C}, \tag{204}$$

the relative uncertainty is

$$\frac{u(L)}{L} = \sqrt{\left(2\frac{u(f_0)}{f_0}\right)^2 + \left(\frac{u(C)}{C}\right)^2}. \tag{205}$$

In the numerical calculation, no separate uncertainty was assigned to $C = 22.0$ nF, so the uncertainty in L comes from the uncertainty in $f_{0,\mathrm{est}}$. This means that the reported uncertainty of L is likely an underestimate, because the fitted capacitance from Tasks 1 and 2 also has systematic uncertainty.

The uncertainty of the mutual inductance was calculated from

$$M = k_L L. \tag{206}$$

Thus,

$$u(M) = \sqrt{(L u(k_L))^2 + (k_L u(L))^2}. \tag{207}$$

The main sources of uncertainty in Task 4 were finite FFT frequency resolution, uncertainty in identifying the two split resonance peaks, overlap of the peaks at larger distances, distance-reading uncertainty of the optical bench, possible misalignment of the two coils, finite coil size and thickness, use of an estimated rather than independently measured f_0 and possible systematic uncertainty in the capacitance C.

7.7 Distance Dependence of the Coupling Factor

Figure 14 shows the inductive coupling factor as a function of coil distance. The coupling factor decreases from

$$k_L = 0.4763 \tag{208}$$

at $d = 4$ cm to

$$k_L = 0.1621 \tag{209}$$

at $d = 18$ cm.

The data were fitted with the power law

$$k_L(d) = Ad^{-n}. \tag{210}$$

The fitted exponent was

$$n = 0.6746. \tag{211}$$

The numerical uncertainty returned by the fit rounded to zero at the displayed precision. This should not be interpreted as a true physical uncertainty of zero; rather, it reflects the numerical fit output and the limited uncertainty model. Systematic effects are expected to dominate the uncertainty of the exponent.

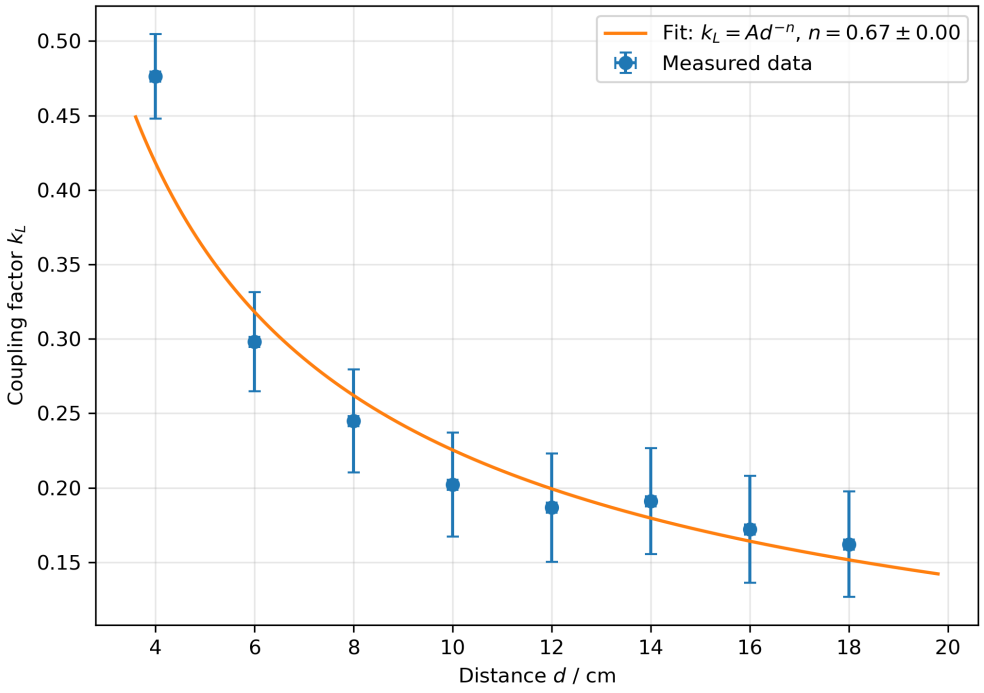


Figure 14: Inductive coupling factor k_L as a function of coil distance d.

The residuals of this fit are shown in Fig. 15. The residuals show that the first point at $d = 4$ cm lies above the fit, while several intermediate points lie below it. This suggests that a simple power law is only an approximate description over the measured distance range.

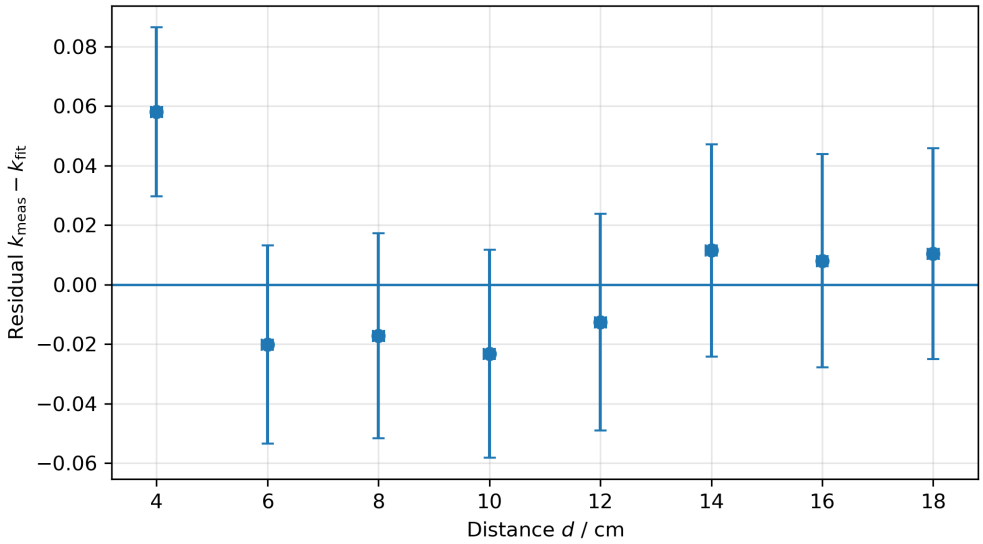


Figure 15: Residuals of the power-law fit to $k_L(d)$, calculated as $k_{\mathrm{meas}} - k_{\mathrm{fit}}$.

7.8 Distance Dependence of the Mutual Inductance

The mutual inductance was calculated using

$$M = k_L L. \tag{212}$$

Since L was treated as constant for all distances, the distance dependence of M follows the same trend as the coupling factor. The mutual inductance decreased from

$$M = 10.2164 \text{ mH} \tag{213}$$

at $d = 4$ cm to

$$M = 3.4774 \text{ mH} \tag{214}$$

at $d = 18$ cm.

Figure 16 shows M as a function of distance together with a power-law fit,

$$M(d) = Ad^{-n}. \tag{215}$$

The fitted exponent was again

$$n = 0.6746. \tag{216}$$

The exponent is the same as for $k_L(d)$ because $M = k_L L$ and L was constant.

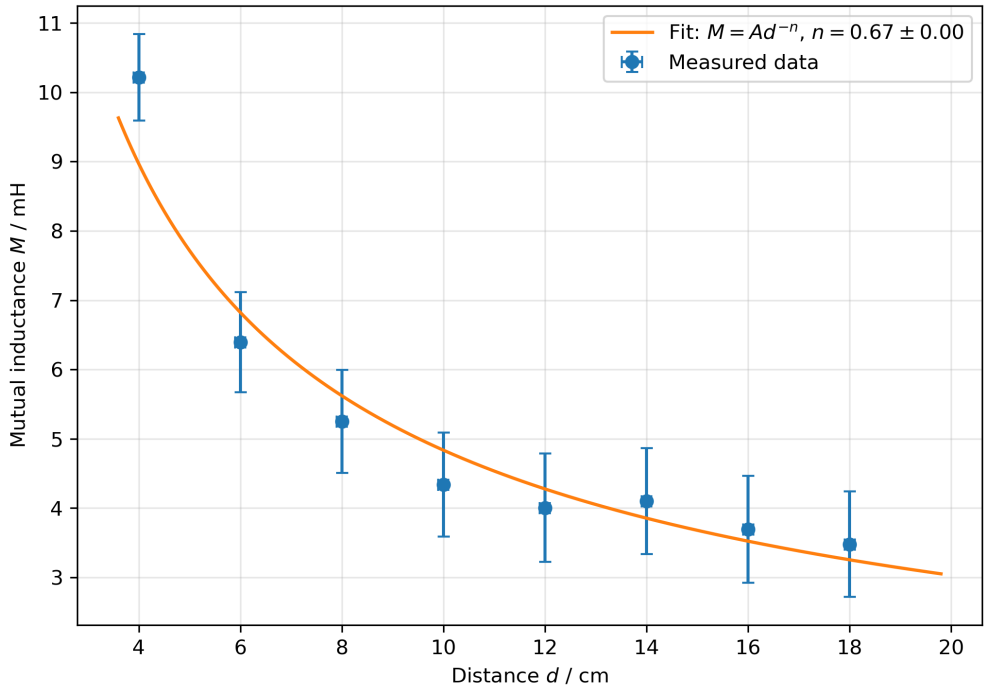


Figure 16: Mutual inductance M as a function of coil distance d.

The same fit is shown on logarithmic axes in Fig. 17. On a log-log plot, a power law appears as a straight line. The approximate linearity confirms that a power-law description is reasonable over the measured range, although the residuals indicate that it is not exact.

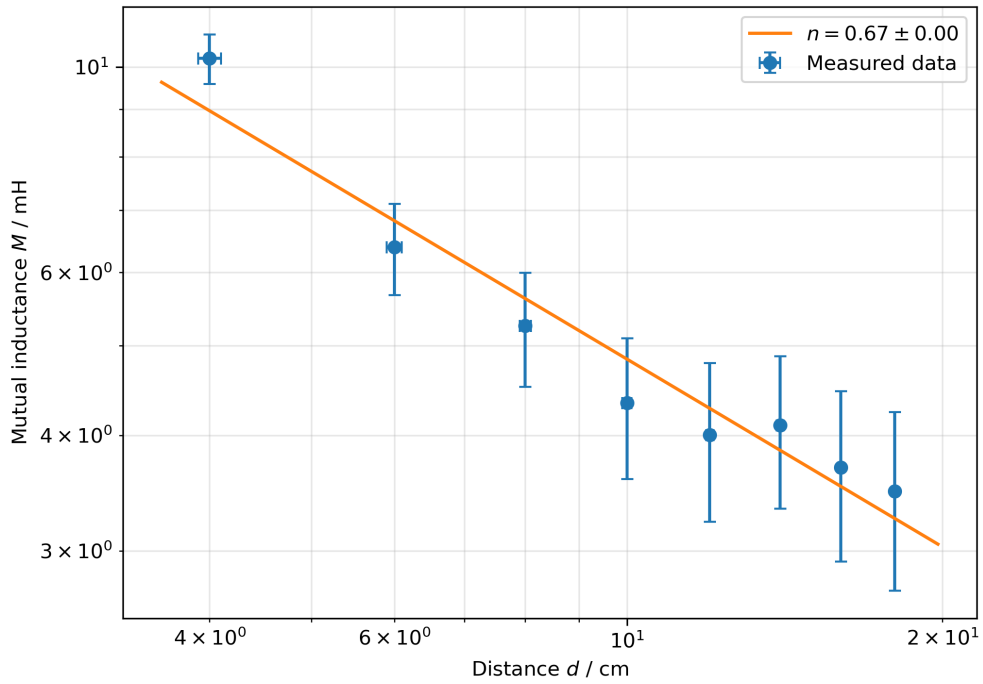


Figure 17: Log-log plot of mutual inductance M versus distance d.

The residuals for the mutual-inductance fit are shown in Fig. 18.

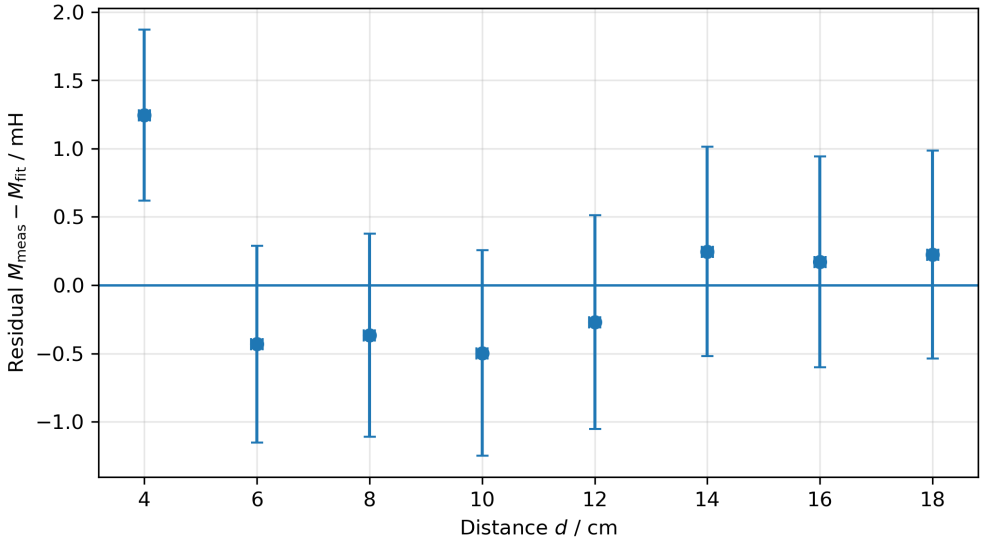


Figure 18: Residuals of the power-law fit to $M(d)$, calculated as $M_{\mathrm{meas}} - M_{\mathrm{fit}}$.

7.9 Interpretation and Discussion

The measured spectra show a clear frequency splitting for the inductively coupled resonant circuits. The splitting is largest at the smallest coil distance and decreases as the distance increases. This confirms the expected physical behaviour: the magnetic flux from one coil through the other decreases with distance, so the mutual inductance and coupling factor decrease.

The coupling factor decreased from

$$k_L = 0.4763 \tag{217}$$

at $d = 4$ cm to

$$k_L = 0.1621 \tag{218}$$

at $d = 18$ cm. The mutual inductance decreased from

$$M = 10.2164 \text{ mH} \tag{219}$$

to

$$M = 3.4774 \text{ mH} \tag{220}$$

over the same distance range.

The fitted power-law exponent was

$$n = 0.6746. \tag{221}$$

This is much smaller than the far-field value $n = 3$ expected from the axial magnetic field of a circular coil at distances much larger than the coil radius. Therefore, the measured distance dependence is not compatible with a simple far-field d^{-3} law.

There are several possible reasons for this. First, the measured distances may not have been large enough compared with the coil radius, so the far-field approximation was probably not valid. In the near-field region, the magnetic field of a circular coil does not follow a pure d^{-3} dependence. Second, finite coil size, finite coil thickness, and coil misalignment can significantly affect the magnetic flux through the second coil. Third, the peak splitting becomes smaller at larger distances, making the extraction of f_+ and f_- more uncertain. Finally, because no uncoupled resonance frequency was measured independently, the values of L and M rely on an estimated f_0, which introduces an additional systematic limitation.

Despite these limitations, the main qualitative result is clear: the inductive coupling decreases as the distance between the coils increases. The coupling-factor plot and mutual-inductance plot both show this trend. The power-law fit provides an empirical description of the measured data, but the exponent should not be interpreted as a precise far-field exponent because the assumptions required for that interpretation were not fully satisfied.

Overall, Task 4 confirms the expected decrease of inductive coupling with coil separation and provides an estimate of the mutual inductance as a function of distance. The analysis also shows that an independent measurement of the uncoupled resonance frequency f_0 would have improved the determination of L and M.

8 Discussion

8.1 Overview of the Experimental Results

The experiment investigated coupled electrical oscillations in capacitively and inductively coupled LC circuits. In all parts of the experiment, the coupling between two resonant circuits caused a splitting of the resonance into two normal modes. These two modes were observed as two peaks in the measured frequency spectra. The measured peak frequencies were then used to calculate the corresponding coupling factors.

For the capacitively coupled circuits, two configurations were studied: the low-point circuit and the high-point circuit. In Task 1, the low-point coupling factor decreased with increasing coupling capacitance C_K, as expected from

$$k_{C,T} = \frac{C}{C + C_K}. \tag{222}$$

In Task 2, the high-point coupling factor increased with increasing C_K, as expected from

$$k_{C,H} = \frac{C_K}{C + C_K}. \tag{223}$$

Both fits gave the same capacitance value,

$$C = 22.000 \text{ nF}. \tag{224}$$

This agreement supports the theoretical description of the two capacitive coupling configurations. However, the fitted uncertainty was printed as zero at the displayed precision, so the result should not be interpreted as having zero experimental uncertainty. The actual uncertainty is dominated by systematic effects such as component tolerances, finite FFT resolution, damping, and peak selection.

In Task 3, a beat oscillation was observed in the high-point circuit. The beat period measured directly from the time trace was

$$T_{S,\text{direct}} = (0.68592 \pm 0.06339) \text{ ms}, \tag{225}$$

while the value calculated from the spectral peak separation was

$$T_{S,\text{calc}} = (0.65536 \pm 0.11585) \text{ ms}. \tag{226}$$

These two values agree within their uncertainties. This confirms the expected relation

$$T_S = \frac{1}{|f_- - f_+|}. \tag{227}$$

In Task 4, the inductive coupling factor decreased as the distance between the coils increased. The coupling factor decreased from

$$k_L = 0.4763 \tag{228}$$

at $d = 4$ cm to

$$k_L = 0.1621 \tag{229}$$

at $d = 18$ cm. The corresponding mutual inductance decreased from

$$M = 10.2164 \text{ mH} \tag{230}$$

to

$$M = 3.4774 \text{ mH}. \tag{231}$$

The measured distance dependence was fitted with a power law,

$$M(d) = Ad^{-n}, \tag{232}$$

giving

$$n = 0.6746. \tag{233}$$

This exponent is much smaller than the far-field value $n = 3$ expected for the axial magnetic field of a circular coil at distances much larger than the coil radius. Therefore, the measured range was most likely not in the far-field regime.

8.2 Comparison of Low-Point and High-Point Capacitive Coupling

The low-point and high-point circuits showed opposite dependences of the coupling factor on the coupling capacitance C_K. This is one of the main physical results of the experiment.

For the low-point circuit, the measured coupling factor increased when C_K was decreased. The measured values ranged from

$$k_{C,T} = 0.0807 \tag{234}$$

at $C_K = 0.22\ \mu\text{F}$ to

$$k_{C,T} = 0.3470 \tag{235}$$

at $C_K = 0.04\ \mu\text{F}$. This agrees with the theoretical expression

$$k_{C,T} = \frac{C}{C + C_K}, \tag{236}$$

because increasing C_K increases the denominator and therefore reduces the coupling factor.

For the high-point circuit, the opposite trend was observed. The measured coupling factor increased from

$$k_{C,H} = 0.3846 \tag{237}$$

at $C_K = 0.01\ \mu\text{F}$ to

$$k_{C,H} = 0.9168 \tag{238}$$

at $C_K = 0.20\ \mu\text{F}$. This agrees with

$$k_{C,H} = \frac{C_K}{C + C_K}, \tag{239}$$

because increasing C_K increases the numerator and therefore increases the coupling factor.

Both capacitive fits gave

$$C = 22.000\ \text{nF}. \tag{240}$$

Since no nominal value of the fixed capacitors was available, a direct comparison with a manufacturer value could not be made. Nevertheless, the agreement between the two independent circuit configurations indicates that the measured frequency splittings are consistent with the theoretical model.

8.3 Beat Oscillation and Frequency Splitting

Task 3 demonstrated the connection between the time-domain and frequency-domain descriptions of coupled oscillations. In the time trace, the signal showed a fast oscillation with a slowly varying amplitude envelope. In the frequency spectrum, this corresponded to two normal-mode peaks at

$$f_+ = (7.6294 \pm 0.1907)\ \text{kHz}, \tag{241}$$

and

$$f_- = (9.1553 \pm 0.1907)\ \text{kHz}. \tag{242}$$

The beat period calculated from these frequencies was compatible with the beat period measured directly from the amplitude envelope. This verifies that the beat frequency is determined by the splitting of the two normal-mode frequencies. The relative difference between the directly measured and calculated beat periods was only

$$4.66\%. \tag{243}$$

The normalized deviation was

$$z = 0.23, \tag{244}$$

which shows that the two results agree well within uncertainty.

The automatic extraction of the fast oscillation period from zero crossings, however, produced an unrealistic value. The direct result was

$$T_{\text{direct}} = (0.00064 \pm 0.00045)\ \text{ms}, \tag{245}$$

while the value calculated from the spectrum was

$$T_{\text{calc}} = (0.11916 \pm 0.00191)\ \text{ms}. \tag{246}$$

The direct value is not physically plausible because it would correspond to a frequency much higher than the observed resonance frequencies. This indicates that the zero-crossing algorithm detected noise or numerical crossings instead of full oscillation periods. Therefore, for the fast oscillation period, the frequency-domain value is more reliable. A manual cursor measurement over several visible oscillations would improve this part of the analysis.

8.4 Distance Dependence of Inductive Coupling

The inductive-coupling measurements showed the expected qualitative behaviour: the coupling factor and mutual inductance decreased as the distance between the coils increased. This occurs because the magnetic flux from one coil through the other decreases with increasing coil separation.

The fitted power-law exponent was

$$n = 0.6746. \tag{247}$$

This value is much smaller than the far-field expectation of approximately $n = 3$. This difference does not necessarily mean that the measurement is wrong. The d^{-3} dependence is expected only when the distance is much larger than the coil radius. In the present experiment, the coil distances were likely comparable to the coil radius, so the measurement was performed in the near-field or intermediate-field region. In this region, the magnetic field of a circular coil is not described by a simple d^{-3} law.

Additional effects may also have influenced the distance dependence. The coils had finite size and thickness, and small misalignments between the coil axes would reduce the magnetic flux through the second coil. At larger distances, the spectral peaks became closer together, which increased the relative uncertainty in the frequency splitting. Since the coupling factor is calculated from the difference between the squared frequencies, small errors in peak identification can noticeably affect k_L when the splitting is small.

Another important limitation is that the uncoupled resonance frequency f_0 was not measured separately. Instead, it was estimated from the two split frequencies using

$$f_0^2 = \frac{2f_+^2 f_-^2}{f_+^2 + f_-^2}. \tag{248}$$

This allowed the self-inductance and mutual inductance to be estimated, but it introduces an additional systematic uncertainty. An independent measurement of f_0 would have made the determination of L and M more reliable.

8.5 Uncertainty and Limitations

The main uncertainty source in all frequency-domain measurements was the finite FFT frequency resolution. For Tasks 2–4, the frequency spacing was

$$\Delta f = 0.381470 \text{ kHz}, \tag{249}$$

so the frequency uncertainty was taken as

$$u(f) = 0.190735 \text{ kHz}. \tag{250}$$

For Task 1, the frequency resolution was better:

$$\Delta f = 0.095367 \text{ kHz}, \tag{251}$$

giving

$$u(f) = 0.047684 \text{ kHz}. \tag{252}$$

The coupling factors were calculated from squared frequency differences. Therefore, their uncertainties depend strongly on how accurately the peak positions can be determined. The uncertainty becomes especially important when the two peaks are close together, as in the inductive-coupling measurements at larger distances.

The capacitance uncertainty of the capacitor decade was taken as

$$u(C_K) = 0.05 C_K. \tag{253}$$

This uncertainty was included in the fits for Tasks 1 and 2. For Task 4, the coil-distance uncertainty was taken as

$$u(d) = 0.1 \text{ cm}. \tag{254}$$

Several systematic uncertainties are not fully captured by the numerical error propagation. These include damping and finite peak width in the spectra, uncertainty in selecting the correct peak positions, non-identical inductors and capacitors in the two resonant circuits, parasitic capacitances and resistances in wires and connections, limited accuracy of the capacitor decade, coil misalignment and finite coil dimensions, absence of an independent uncoupled-frequency measurement in Task 4 and limitations of automatic period extraction in Task 3.

For these reasons, the numerical fit uncertainties, especially those printed as zero, should not be interpreted as the true total experimental uncertainty. They only represent the uncertainty within the simplified fitting model.

8.6 Overall Evaluation

Overall, the experiment successfully demonstrated the key features of coupled electrical oscillations. In both capacitively and inductively coupled circuits, the coupling caused a splitting of the resonance into two normal modes. The measured frequency spectra allowed the normal-mode frequencies to be determined and the coupling factors to be calculated.

The low-point and high-point capacitive measurements confirmed the two different theoretical dependences of the coupling factor on C_K. Both analyses gave a consistent capacitance value of

$$C = 22.000 \text{ nF}. \tag{255}$$

The beat measurement confirmed the relationship between the beat period and the frequency difference of the two normal modes. The inductive-coupling measurement showed that the coupling factor and mutual inductance decrease with coil distance, although the measured distance dependence was not compatible with a far-field d^{-3} law.

The most important limitations were the finite FFT resolution, uncertainty in peak selection, the lack of a nominal capacitor value for comparison, the failed automatic extraction of the fast oscillation period in Task 3, and the absence of an independently measured uncoupled resonance frequency in Task 4. Despite these limitations, the main physical behaviour predicted by the theory was observed in all four tasks.

9 Conclusion

In this experiment, coupled electrical oscillations were investigated using capacitively and inductively coupled LC resonant circuits. The measured time traces and Fourier spectra showed the expected splitting of the resonance into two normal modes. These normal-mode frequencies were used to calculate the coupling factors for the different circuit configurations.

For the low-point capacitively coupled circuit, the coupling factor increased as the coupling capacitance C_K decreased, in agreement with

$$k_{C,T} = \frac{C}{C + C_K}. \tag{256}$$

For the high-point capacitively coupled circuit, the opposite trend was observed: the coupling factor increased with increasing C_K, as predicted by

$$k_{C,H} = \frac{C_K}{C + C_K}. \tag{257}$$

Both fits gave the same capacitance value,

$$C = 22.000 \text{ nF}. \tag{258}$$

This agreement confirms the consistency of the capacitive-coupling model.

The beat oscillation measurement demonstrated the relationship between the time-domain amplitude modulation and the frequency-domain mode splitting. The directly measured beat period,

$$T_{S,\text{direct}} = (0.68592 \pm 0.06339) \text{ ms}, \tag{259}$$

agreed within uncertainty with the value calculated from the two spectral peak frequencies,

$$T_{S,\text{calc}} = (0.65536 \pm 0.11585) \text{ ms}. \tag{260}$$

For the inductively coupled circuits, the coupling factor and mutual inductance decreased as the coil distance increased. The mutual inductance decreased from

$$M = 10.2164 \text{ mH} \tag{261}$$

at $d = 4$ cm to

$$M = 3.4774 \text{ mH} \tag{262}$$

at $d = 18$ cm. A power-law fit gave

$$M(d) \propto d^{-0.6746}. \tag{263}$$

This exponent is much smaller than the far-field value expected for the axial magnetic field of a circular coil, indicating that the measured distances were not in the far-field regime and that finite coil geometry and alignment effects were significant.

Overall, the experiment confirmed the main theoretical predictions for coupled electrical oscillations: resonance splitting, normal modes, beat formation, and decreasing inductive coupling with distance. The largest limitations were the finite FFT resolution, peak-selection uncertainty, damping, component tolerances, and the absence of an independently measured uncoupled resonance frequency in Task 4.

10 References

M. Ziese, E17e Fourier Analysis of Coupled Electric Oscillations, University of Leipzig, May 2026.